

Project Data Page:





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Nozzle Calcs.: K9-2"	145



Process Air 2nd KOD 197-D

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Process Air Compressor 2nd KO Drum-197-D-7%.pvdb

Date of Analysis : Sep 16, 2026 2:27pm

PV Elite 28 U1, March 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 3806.000 mm. ID
 Vessel Design Length, Tangent to Tangent 3800.00 mm.
 Distance of Bottom Tangent above Grade 3000.00 mm.
 Distance of Base above Grade 300.00 mm.
 Specified Datum Line Distance 2700.00 mm.
 Internal Design Temperature 82 °C
 Internal Design Pressure 10.000 bars
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 6 bars
 Shop Test Pressure 13.870 bars
 Field Test Pressure 13.000 bars
 Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature -17.0 °C
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Skirt	SA-36	...	...	K02600	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-312 TP316L	...	...	S31603	
Nozzle	SA-182 F316L	...	<= 130	S31603	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-105	...	...	K03504	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-182 F316L	...	<= 130	S31603	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-516 70	...	...	K02700	
Base Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt2	...	...	...	1.0000	N/A	No
Skirt1	...	...	...	1.0000	N/A	No
Bottom Clad Dis	10.156	0.00	10.000	1.0000	No	No
Clad shell 1	10.058	0.00	10.000	1.0000	N/A	No
Unclad Shell	10.000	0.00	10.000	1.0000	N/A	No
Unclad Topdish	10.000	0.00	10.000	1.0000	No	No

Liquid Level: 1601.50 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.991

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
---------	-----------	---------	---------	----------	----------	----------	------	------



Type	mm.	Length mm.	Thickness mm.	Thickness mm.	Internal mm.	External mm.	Eff	Eff
Skirt	-1000.0	1700.0	12.0	12.0	...	...	0.70	0.70
Skirt	0.0	1000.0	10.0	10.0	...	...	0.70	0.50
Ellipse	50.0	50.0	22.0	18.0	17.5	...	0.85	0.85
Cylinder	1050.0	1000.0	18.0	18.0	17.4	...	0.85	0.85
Cylinder	3750.0	2700.0	18.0	18.0	17.3	...	0.85	0.85
Ellipse	3800.0	50.0	22.0	18.0	17.2	...	0.85	0.85

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Lift Lug	WRC-107/537	0.840	n/a	Passed
D1-2"	WRC-107/537	0.450	n/a	Passed
N1-30"	WRC-297	1.049	Nozzle	Failed
N2-30"	WRC-297	1.049	Nozzle	Failed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	2344.96	8722.71	10702	42247.9
20	30	2200	2029.71	8593.46	6983.5	27528.9
30	40	2725	1844.37	8497.89	5046.42	18983.1
40	50	3250	1579.42	6566.73	4960.82	18606.4
50	60	5100	1393.75	5298	3474.2	12674
60	70	6475	892.414	3093.96	387.828	1344.58

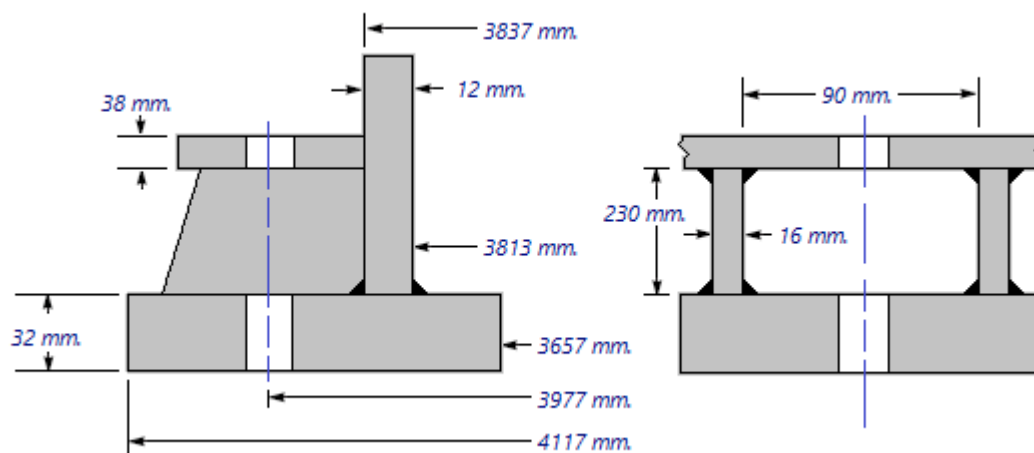
Abs Max of the all of the Stress Ratio's : 0.4714

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	32.0000	mm.
Inside Diameter of Basing	3657.0000	mm.
Outside Diameter of Basing	4117.0000	mm.
Nominal Diameter of Bolts	42.0000	mm.
Diameter of Bolt Circle	3976.9998	mm.
Number of Bolts	20	

Thickness of Gusset Plates	16.0000	mm.
Average Width of Gusset Plates	130.0000	mm.
Height of Gussets	230.0000	mm.
Distance between Gussets	90.0000	mm.
Thickness of Top Plate or Ring	38.0000	mm.
Radial Width of the Top Plate	140.0000	mm.
Number of Gussets	40	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	2345. Kg
Total Earthquake Shear on Support	8723. Kg
Wind Moment on Support	11092. Kg-m.
Earthquake Moment on Support	42638. Kg-m.

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	24329.2 kg.
Shop Test - Fabricated + water (full)	82220.5 kg.
Shipping - Fab. + rem. intls.+ shipping app.	24329.2 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	36155.1 kg.
Empty - Fab. + intls. + details + wghts.	36155.1 kg.
Operating - Empty + operating liquid (no ca)	50572.3 kg.
Field Test - Empty weight + test medium (full)	93556.5 kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
D1-2"	6	...	...	OK	3.42	OK	No Calc[*]
K1-2"	6	...	...	OK	4.80	OK	No Calc[*]
K3-2"	6	...	...	OK	4.80	OK	No Calc[*]
K5-2"	6	...	...	OK	4.80	OK	No Calc[*]
K7-2"	6	...	...	OK	4.80	OK	No Calc[*]
K2-2"	6	...	...	OK	4.80	OK	No Calc[*]
K4-2"	6	...	...	OK	4.80	OK	No Calc[*]
K6-2"	6	...	...	OK	4.80	OK	No Calc[*]
K8-2"	6	...	...	OK	4.80	OK	No Calc[*]
K10-2"	6	...	...	OK	4.80	OK	No Calc[*]
N1-30"	6	...	...	OK	9.30	OK	Passed
N2-30"	6	...	...	OK	9.30	OK	Passed
M1-24"	6	...	...	...	...	OK	Passed
K9-2"	6	...	...	OK	5.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6 Nozzle : K9-2"
Minimum MAWP Shells/Flanges : 6 Element : Bottom Clad Dis
Minimum MAPnc Shells/Flanges : 11 Element : Unclad Shell

Computed Vessel M.A.W.P. : 6 bars

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	D1-2"	0.000	0.000	101.270
40	K1-2"	850.000	90.000	118.328
40	K3-2"	850.000	80.000	118.328
40	K5-2"	850.000	100.000	118.328
40	K7-2"	850.000	270.000	118.328
40	K2-2"	150.000	80.000	118.328
40	K4-2"	150.000	90.000	118.328
40	K6-2"	150.000	100.000	118.328
40	K8-2"	150.000	110.000	118.328
40	K10-2"	150.000	270.000	118.328
50	N1-30"	1850.000	90.000	1464.000
50	N2-30"	2450.001	270.000	1464.000
50	M1-24"	1600.000	180.000	1191.200
50	K9-2"	2600.000	110.000	118.328

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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**Nozzle Schedule:**

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1-2"	2.000 in	80S	WNF	2.375	5.537	...	...	168.13	600
K1-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	224.84	300
K3-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	224.84	300
K5-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	224.84	300
K7-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	224.84	300
K2-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	224.84	300
K4-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	224.84	300
K6-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	224.84	300
K8-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	224.84	300
K10-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	224.84	300
K9-2"	2.000 in	Actual	LWN	3.320	16.764	...	...	224.84	300
M1-24"	24.000 in	Actual	WNF	24.000	10.000	1009.60	20.00	342.57	300
N1-30"	30.000 in	Actual	WNF	30.000	18.000	1262.00	20.00	356.53	150
N2-30"	30.000 in	Actual	WNF	30.000	18.000	1262.00	20.00	356.53	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1-2"	SA-312 TP316L	18.000	10.000	...	...	...
K1-2"	SA-182 F316L	30.000	10.000	...	...	...
K3-2"	SA-182 F316L	30.000	10.000	...	...	...
K5-2"	SA-182 F316L	18.000	10.000	...	...	...
K7-2"	SA-182 F316L	30.000	10.000	...	...	...
K2-2"	SA-182 F316L	30.000	10.000	...	...	...
K4-2"	SA-182 F316L	30.000	10.000	...	...	...
K6-2"	SA-182 F316L	30.000	10.000	...	...	...
K8-2"	SA-182 F316L	30.000	10.000	...	...	...
K10-2"	SA-182 F316L	30.000	10.000	...	...	...
K9-2"	SA-105	30.000	10.000	...	...	...
M1-24"	SA-516 70	18.000	10.000	18.000	20.000	...
N1-30"	SA-516 70	18.000	10.000	18.000	20.000	...
N2-30"	SA-516 70	18.000	10.000	18.000	20.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
D1-2"	...	0.0	150.00	0.00	Bottom Clad Dis
K1-2"	850.000	90.0	206.38	0.00	Clad shell 1
K3-2"	850.000	80.0	206.38	0.00	Clad shell 1
K5-2"	850.000	100.0	206.38	0.00	Clad shell 1
K7-2"	850.000	270.0	206.38	0.00	Clad shell 1
K2-2"	150.000	80.0	206.38	0.00	Clad shell 1
K4-2"	150.000	90.0	206.38	0.00	Clad shell 1
K6-2"	150.000	100.0	206.38	0.00	Clad shell 1

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User



FileName : Process Air Compressor 2nd KO Drum-197-D-7%

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Nozzle Schedule:

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HEXAGON

K8-2"	150.000	110.0	206.38	0.00	Clad shell 1
K10-2"	150.000	270.0	206.38	0.00	Clad shell 1
K9-2"	2600.000	110.0	206.38	0.00	Unclad Shell
M1-24"	1600.000	180.0	300.00	0.00	Unclad Shell
N1-30"	1850.000	90.0	300.00	0.00	Unclad Shell
N2-30"	2450.001	270.0	300.00	0.00	Unclad Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 22.0mm. THK X 3806.0mm. ID X 50.0mm.	SA-516 70
1	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 2700.0mm	...
2	CLADDING: 1001.500mm X 3.000mm THK	...
1	CLADDING: 1000.000mm X 3.000mm THK	...
1	CLADDING: 2700.000mm X 3.000mm THK	...
2	LIFTING LUGS:	SA-516 70
2	LIFTING LUG PADS: 500.0mm X 480.0mm X 20.0mm THK	
1	PLATFORM: 1000mm X 6000mm WIDE	
1	LADDER: 4800.0mm	...
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 4117.0mm. OD X 3657.0 ID X 32.0mm.	SA-516 70
20	BASERING BOLTS: 42.0mm. DIA	SA-193 B7
20	BASERING BOLT NUTS: 42.0mm. DIA	...
1	BASERING TOP RING: 4117.0mm. OD X 3837.0 ID X 38.0mm.	...
40	BASERING GUSSET PLATES: 130.0mm. X 130.0mm. X 230.0mm. X 16.0mm.	...
2	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 3936.3mm. X 1700.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 1968.1mm. X 1700.0mm.	SA-36
1	CYLINDRICAL SKIRT SEGMENT 12.0mm. THK X 2138.0mm. X 1700.0mm.	SA-36
2	CYLINDRICAL SKIRT SEGMENT 10.0mm. THK X 3938.3mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 10.0mm. THK X 1969.2mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 10.0mm. THK X 2139.1mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 18.0mm. THK X 3929.0mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm. THK X 1964.5mm. X 1000.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm. THK X 2134.1mm. X 1000.0mm.	SA-516 70
2	CYLINDRICAL SEGMENT 18.0mm. THK X 3929.1mm. X 2700.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm. THK X 1964.5mm. X 2700.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 18.0mm. THK X 2134.1mm. X 2700.0mm.	SA-516 70
1	NAMEPLATE	...

**Minimum Design Metal Temperature Results Summary (°C):**

Description Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Bottom Clad D[10]	B	-11	-45	-29	0.487	18.000	0.85	No
Bottom Clad Di[7]	B	-5	-48	-29	0.395	22.000	0.85	No
Clad shell 1 [8]	B	-11	-44	-29	0.488	18.000	0.85	No
Unclad Shell [8]	B	-11	-44	-29	0.488	18.000	0.85	No
Unclad Topdis[10]	B	-11	-45	-29	0.487	18.000	0.85	No
Unclad Topdish[7]	B	-5	-48	-29	0.395	22.000	0.85	No
N1-30" [1]	B	-11	-44	-29	0.488	18.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
N2-30" [1]	B	-11	-44	-29	0.488	18.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
M1-24" [1]	B	-11	-44	-29	0.488	18.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
K9-2" [1]	B	-13	-104	-29	0.007	16.764	1.00	No
Nozzle Flg [5]	!	-17	-17					
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Warmest MDMT:		-5	-17					

Required Minimum Design Metal Temperature -12.0 °C

Warmest Computed Minimum Design Metal Temperature -17.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Process Air 2nd KOD 197-D

Design Internal Pressure (for Hydrotest) 10 bars
 Design Internal Temperature 82.0 °C
 Projection of Nozzle from Vessel Top 500 mm.
 Projection of Nozzle from Vessel Bottom 500 mm.
 Minimum Design Metal Temperature -12.0 °C
 Type of Construction Welded
 Special Service None
 Degree of Radiography RT-3
 Use Higher Longitudinal Stresses (Flag) Y
 Select t for Internal Pressure (Flag) N
 Select t for External Pressure (Flag) N
 Select t for Axial Stress (Flag) N
 Select Location for Stiff. Rings (Flag) N
 Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test UG-99(c)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1 NP+EW+1.1WI+FW+BW
 Load Case 2 NP+EW+EE+FS+BS
 Load Case 3 NP+OW+1.1WI+FW+BW
 Load Case 4 NP+OW+EQ+FS+BS
 Load Case 5 NP+HW+HI
 Load Case 6 NP+HW+HE
 Load Case 7 IP+OW+1.1WI+FW+BW
 Load Case 8 IP+OW+EQ+FS+BS
 Load Case 9 EP+OW+1.1WI+FW+BW
 Load Case 10 EP+OW+EQ+FS+BS
 Load Case 11 HP+HW+HI
 Load Case 12 HP+HW+HE



Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.050
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

**Complete Listing of Vessel Elements and Details:**

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt2
Distance "FROM" to "TO"	1700 mm.
Skirt Outside Diameter	3837 mm.
Diameter of Skirt at Base	3837 mm.
Skirt Thickness	12 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	12 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.28
Material Name	SA-36
Allowable Stress, Ambient	114 N./mm2
Allowable Stress, Operating	114 N./mm2
Allowable Stress, Hydrotest	225 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	230.08 N./mm2
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K02600
Product Form	Plate; sheet
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/6
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt1
Distance "FROM" to "TO"	1000 mm.
Skirt Outside Diameter	3835 mm.
Diameter of Skirt at Base	3835 mm.
Skirt Thickness	10 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	10 mm.
External Corrosion Allowance	0 mm.
Design Temperature Internal Pressure	82 °C
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.28
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	242.6 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.5
Weld is pre-Heated	No

Element#	3/6
Element From Node	30
Element To Node	40



Element Type	Elliptical
Description	Bottom Clad Dis
Distance "FROM" to "TO"	50 mm.
Inside Diameter	3806 mm.
Element Thickness	18 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	22 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.28
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	0 mm.
Platform Start Angle (degrees)	360.0
Platform End Angle (degrees)	0.0
Platform Wind Area	30000 cm2
Platform Weight	4517.9 Kg
Platform Railing Weight	0 Kg/mm.
Platform Grating Weight	0.017 Kgs/cm2
Platform Width	1500 mm.
Platform Height	1000 mm.
Platform Clearance or End Offset	85 mm.
Platform Force Coefficient	1.0
Ladder Layout Angle	90.0
Ladder Start Elevation	0 mm.
Ladder End Elevation	2700 mm.
Unit Weight of Ladder	0.037 Kg/mm.
Platform Length (top head platform)	0 mm.

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	-951.5 mm.
Height/Length of Liquid	1001.5 mm.
Liquid Density	0.0009894 kg./cm3

Element From Node	30
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	-951.5 mm.
Height/Length of Lining	1001.5 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm3

Element From Node	30
Detail Type	Nozzle
Detail ID	D1-2"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	80S
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-312 TP316L



Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Clad shell 1
Distance "FROM" to "TO"	1000 mm.
Inside Diameter	3806 mm.
Element Thickness	18 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	18 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.28
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	600 mm.
Liquid Density	0.0009894 kg./cm3
Element From Node	40
Detail Type	Lining
Detail ID	clad
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	1000 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm3
Element From Node	40
Detail Type	Nozzle
Detail ID	K1-2"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L
Element From Node	40
Detail Type	Nozzle
Detail ID	K3-2"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	80.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L
Element From Node	40
Detail Type	Nozzle
Detail ID	K5-2"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300



Layout Angle	100.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K7-2"	
Dist. from "FROM" Node / Offset dist	800	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K2-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	80.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K4-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K6-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	100.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	
Nozzle Matl	SA-182 F316L	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	K8-2"	
Dist. from "FROM" Node / Offset dist	100	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	110.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 2.3	



Nozzle Matl	SA-182 F316L
Element From Node	40
Detail Type	Nozzle
Detail ID	K10-2"
Dist. from "FROM" Node / Offset dist	100 mm.
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 2.3
Nozzle Matl	SA-182 F316L

Element From Node	40
Detail Type	Weight
Detail ID	nozzles lining
Dist. from "FROM" Node / Offset dist	10 mm.
Miscellaneous Weight	30 Kg
Offset from Element Centerline	0 mm.

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Unclad Shell
Distance "FROM" to "TO"	2700 mm.
Inside Diameter	3806 mm.
Element Thickness	18 mm.
Internal Corrosion Allowance	1 mm.
Nominal Thickness	18 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	10 bars
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.28
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	50
Detail Type	Lining
Detail ID	lining
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Lining	2700 mm.
Thickness of Lining	3 mm.
Density	0.008 kg./cm3

Element From Node	50
Detail Type	Nozzle
Detail ID	N1-30"
Dist. from "FROM" Node / Offset dist	800 mm.
Nozzle Diameter	30 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element From Node	50
Detail Type	Nozzle
Detail ID	N2-30"



Dist. from "FROM" Node / Offset dist	1400	mm.
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	M1-24"	
Dist. from "FROM" Node / Offset dist	550	mm.
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	K9-2"	
Dist. from "FROM" Node / Offset dist	1550	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	110.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	

Element From Node	50	
Detail Type	Weight	
Detail ID	Donut Ring	
Dist. from "FROM" Node / Offset dist	-0.0001463	mm.
Miscellaneous Weight	160	Kgf
Offset from Element Centerline	0	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	Demister pad	
Dist. from "FROM" Node / Offset dist	-0.0001463	mm.
Miscellaneous Weight	250	Kgf
Offset from Element Centerline	0	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	N1 Distributor	
Dist. from "FROM" Node / Offset dist	1000	mm.
Miscellaneous Weight	1250	Kgf
Offset from Element Centerline	1000	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	N2-Hopper	
Dist. from "FROM" Node / Offset dist	1400	mm.
Miscellaneous Weight	860	Kgf
Offset from Element Centerline	1000	mm.

Element From Node	50	
Detail Type	Weight	
Detail ID	Davit	
Dist. from "FROM" Node / Offset dist	550	mm.
Miscellaneous Weight	50	Kgf



Offset from Element Centerline 1500 mm.

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	Unclad Topdish	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	3806	mm.
Element Thickness	18	mm.
Internal Corrosion Allowance	1	mm.
Nominal Thickness	22	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	10	bars
Design Temperature Internal Pressure	82	°C
Design External Pressure	0	bars
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.28	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	1050	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	79507	cm2
Platform Weight	6300.7	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	6000	mm.
Platform Height	1000	mm.
Platform Clearance or End Offset	3000	mm.
Platform Force Coefficient	1.0	
Ladder Layout Angle	135.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	4800	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	6000	mm.
Element From Node	60	
Detail Type	Lining	
Detail ID	clad	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Lining	1001.5	mm.
Thickness of Lining	3	mm.
Density	0.008	kg./cm3
Element From Node	60	
Detail Type	Weight	
Detail ID	Clip weight	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	480	Kgf
Offset from Element Centerline	0	mm.

**XY Coordinate Calculations:**

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
	Skirt2	...	1700	...	1700
	Skirt1	...	2700	...	1000
	Bottom Clad Dis	...	2750	...	50
	Clad shell 1	...	3750	...	1000
	Unclad Shell	...	6450	...	2700
	Unclad Topdish	...	6500	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Skirt2		...	12	1	3837	...	1
Skirt1		...	10	1	3835	...	2
Bottom Clad Dis		10.156	22	1	3806	117.3	3
Clad shell 1		10.058	18	1	3806	117.3	4
Unclad Shell		10	18	1	3806	117.3	5
Unclad Topdish		10	22	1	3806	117.3	6

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Skirt2		...	No Calc	No Calc	12	No Calc	1
Skirt1		...	No Calc	No Calc	10	No Calc	2
Bottom Clad Dis		10	10.315	11.084	18	17.4881	3
Clad shell 1		10	10.3585	11.0319	18	17.4119	4
Unclad Shell		10	10.4168	11.0319	18	17.3163	5
Unclad Topdish		10	10.4706	11.084	18	17.2353	6

Minimum 6 11

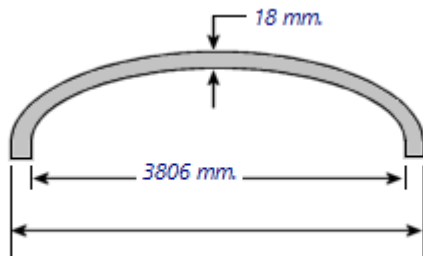
MAWP: 6 bars, limited by: DESIGN (user specified)

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom Clad Dis

Elliptical Head Design Information:					
Design Pressure	P	10.156 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	3806.000 mm.	Actual Stress		133.8 N./mm^2
Spec. Min. Thk t _e , or s _a		18.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		166721.3 cm^2	SG of Contents		0.991
Element Volume		7787050.5 cm3	Weight of Contents		7676.6 Kg
Empty Weight		3092.8 Kg	Operating Weight		10769.4 Kg
			Comp. All. Ope. Stress		106.0 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 47.575 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 23.788 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (10.156 \cdot 3808.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 10.156)$$

$$= 16.4881 + 1.0000 = 17.4881 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

[Less Operating Hydrostatic Head Pressure of 0.156 bars](#)

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 17.0) / (0.999 \cdot 3808.0 + 0.2 \cdot 17.0)$$

$$= 10 - 0 = 10 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 18.0) / (1.0 \cdot 3806.0002 + 0.2 \cdot 18.0)$$

$$= 11 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (10.156 \cdot (0.999 \cdot 3808.0 + 0.2 \cdot 17.0)) / (2 \cdot 0.85 \cdot 17.0)$$

$$= 133.848 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (10.156 \cdot 1904.0) / (138.0 \cdot 0.85 - 0.6 \cdot 10.156) + 1.0$$

$$= 17.571 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.063 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 21.0) / (1904.0 + 0.6 \cdot 21.0)$$

$$= 13 - 0.1 = 13 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (3808.0 / (2 \cdot 952.5)))^2 / 6$$

$$= 0.999300$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o)$ 2.508 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 18.0$, $t_r = 9.738$, $c = 1.0 \text{ mm.}$, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.487$, Temp. Reduction = 34 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-45 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 22.0$, $t_r = 9.77$, $c = 1.0 \text{ mm.}$, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.395$, Temp. Reduction = 52 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.395	Temperature Difference	44 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

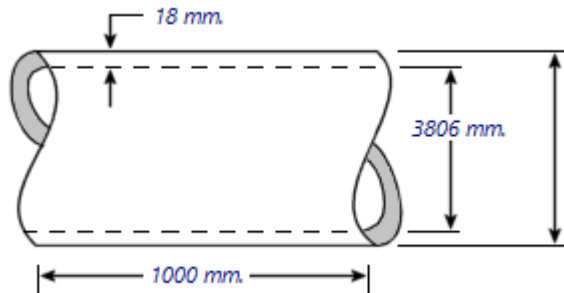
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 47.575 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 23.788 mm.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Clad shell 1

Cylindrical Shell Design Information:			
Design Pressure P	10.058 bars	Int. Design Temperature	82 °C
Material	SA-516 70	Ext. Design Temperature	0 °C
Length	1000.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	3806.000 mm.	Actual Stress	133.3 N./mm ²
Spec. Min. Thk t,e, or sa	18.0000 mm.	Spec. Nominal Thk.	18.000 mm.
Surface Area	120700.0 cm ²	SG of Contents	0.991
Element Volume	11379032.0 cm ³	Weight of Contents	6740.5 Kg
Empty Weight	1675.9 Kg	Operating Weight	8416.4 Kg
		Comp. All. Ope. Stress	106.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.058 \cdot 1904.0) / (138.0 \cdot 0.85 - 0.6 \cdot 10.058)$$

$$= 16.4119 + 1.0000 = 17.4119 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.058 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 17.0) / (1904.0 + 0.6 \cdot 17.0)$$

$$= 10 - 0.1 = 10 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 18.0) / (1903.0 + 0.6 \cdot 18.0)$$

$$= 11 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.058 \cdot (1904.0 + 0.6 \cdot 17.0)) / (0.85 \cdot 17.0)$$

$$= 133.250 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.471 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 18.0, tr = 9.77, c = 1.0 mm., E* = 0.85

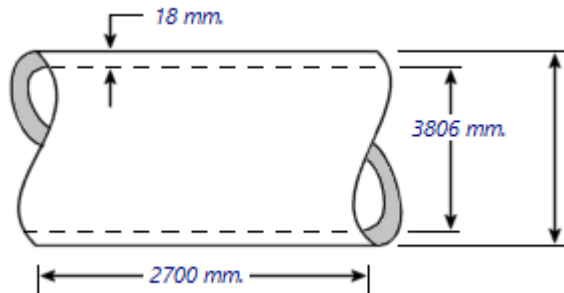
Thickness Ratio = $tr * E^* / (tg - c) = 0.488$, Temp. Reduction = 34 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm.	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.488	Temperature Difference	34 °C
UCS-66(c) red.	No	Computed Min. Temp.	-44 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

**Internal Pressure Calculation Results:****ASME Code, Section VIII Division 1, 2025****Internal Pressure Results for: Unclad Shell**

Cylindrical Shell Design Information:					
Design Pressure	P	10.000 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		2700.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	3806.000 mm.	Actual Stress		132.5 N./mm^2
Spec. Min. Thk t,e, or sa		18.0000 mm.	Spec. Nominal Thk.		18.000 mm.
Surface Area		325890.0 cm^2	SG of Contents		0.000
Element Volume		30723384.0 cm3	Weight of Contents		0.0 Kg
Empty Weight		4525.0 Kg	Operating Weight		4525.0 Kg
			Comp. All. Ope. Stress		106.0 N./mm^2

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (10.0 \cdot 1904.0) / (138.0 \cdot 0.85 - 0.6 \cdot 10.0)$$

$$= 16.3163 + 1.0000 = 17.3163 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 17.0) / (1904.0 + 0.6 \cdot 17.0)$$

$$= 10 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 18.0) / (1903.0 + 0.6 \cdot 18.0)$$

$$= 11 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (10.0 \cdot (1904.0 + 0.6 \cdot 17.0)) / (0.85 \cdot 17.0)$$

$$= 132.478 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.471 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18.0, tr = 9.77, c = 1.0 mm., E* = 0.85



Thickness Ratio = $tr * E / (tg - c) = 0.488$, Temp. Reduction = 34 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C

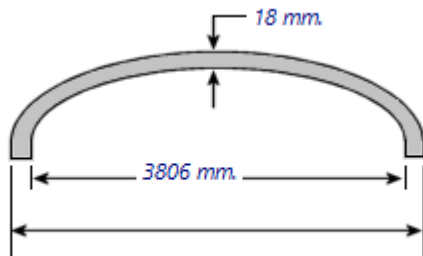
Min Metal Temp. at Required thickness (UCS 66.1) -44 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	18.000 mm.	Basic MDMT	-11 °C
Stress/Thk. Ratio	0.488	Temperature Difference	34 °C
UCS-66(c) red.	No	Computed Min. Temp.	-44 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Unclad Topdish

Elliptical Head Design Information:					
Design Pressure	P	10.000 bars	Int. Design Temperature		82 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	1.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	3806.000 mm.	Actual Stress		131.8 N./mm^2
Spec. Min. Thk t _e , or s _a		18.0000 mm.	Spec. Nominal Thk.		22.000 mm.
Surface Area		166721.3 cm^2	SG of Contents		0.000
Element Volume		7787050.5 cm3	Weight of Contents		0.0 Kgf
Empty Weight		3092.8 Kgf	Operating Weight		3092.8 Kgf
			Comp. All. Ope. Stress		106.0 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 47.575 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 23.788 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (10.0 \cdot 3808.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 10.0)$$

$$= 16.2353 + 1.0000 = 17.2353 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 17.0) / (0.999 \cdot 3808.0 + 0.2 \cdot 17.0)$$

$$= 10 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 18.0) / (1.0 \cdot 3806.0002 + 0.2 \cdot 18.0)$$

$$= 11 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (10.0 \cdot (0.999 \cdot 3808.0 + 0.2 \cdot 17.0)) / (2 \cdot 0.85 \cdot 17.0)$$

$$= 131.798 \text{ N./mm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (10.0 \cdot 1904.0) / (138.0 \cdot 0.85 - 0.6 \cdot 10.0) + 1.0$$

$$= 17.316 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 21.0) / (1904.0 + 0.6 \cdot 21.0)$$

$$= 13 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (3808.0 / (2 \cdot 952.5))^2) / 6$$

$$= 0.999300$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.508 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 18.0, tr = 9.738, c = 1.0 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.487$, Temp. Reduction = 34 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-45 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 22.0, tr = 9.77, c = 1.0 mm., E* = 0.85

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.395$, Temp. Reduction = 52 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.000 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	22.000 mm.	Basic MDMT	-5 °C
Stress/Thk. Ratio	0.395	Temperature Difference	44 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	85 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 47.575 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 23.788 mm.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7.800 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.000 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13.870 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	6.600 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	14.300 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.300 bars
App 27-4	= M.A.W.P.	6.000 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Clad Dis	177.3	234.0	0.758	14.24
Clad shell 1	178.2	234.0	0.761	14.24
Unclad Shell	178.2	234.0	0.761	14.24
Unclad Topdish	177.3	234.0	0.758	14.24

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	115.00	115.00	1.000
K1-2"	Nozzle	115.00	101.29	1.135
K3-2"	Nozzle	115.00	101.29	1.135
K5-2"	Nozzle	115.00	101.29	1.135
K7-2"	Nozzle	115.00	101.29	1.135
K2-2"	Nozzle	115.00	101.29	1.135
K4-2"	Nozzle	115.00	101.29	1.135
K6-2"	Nozzle	115.00	101.29	1.135
K8-2"	Nozzle	115.00	101.29	1.135
K10-2"	Nozzle	115.00	101.29	1.135
N1-30"	Nozzle	138.00	138.00	1.000
N1-30"	Pad	138.00	138.00	1.000
N2-30"	Nozzle	138.00	138.00	1.000
N2-30"	Pad	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K9-2"	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Clad Dis	138.00	138.00	1.000
Clad shell 1	138.00	138.00	1.000
Unclad Shell	138.00	138.00	1.000
Unclad Topdish	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
D1-2"	8.30	153.00	0.054
K1-2"	3.01	153.00	0.020
K3-2"	3.01	153.00	0.020



K5-2"	3.01	153.00	0.020
K7-2"	3.01	153.00	0.020
K2-2"	3.01	153.00	0.020
K4-2"	3.01	153.00	0.020
K6-2"	3.01	153.00	0.020
K8-2"	3.01	153.00	0.020
K10-2"	3.01	153.00	0.020
N1-30"	29.58	234.00	0.126
N2-30"	29.58	234.00	0.126
M1-24"	42.85	234.00	0.183
K9-2"	3.01	225.00	0.013

Elements Suitable for Test Pressure.

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**Hydrostatic Test Pressure Results:**

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	7.800 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	13.000 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	13.929 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	6.600 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	14.300 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	14.300 bars
App 27-4	= M.A.W.P.	6.000 bars

UG-99(b) Note 35, Test Pressure Calculation:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 10.0 • 1.0

= 13.000 bars

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Bottom Clad Dis	179.4	234.0	0.766	13.61
Clad shell 1	179.0	234.0	0.765	13.51
Unclad Shell	177.7	234.0	0.759	13.41
Unclad Topdish	173.3	234.0	0.740	13.15

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
D1-2"	Nozzle	115.00	115.00	1.000
K1-2"	Nozzle	115.00	101.29	1.135
K3-2"	Nozzle	115.00	101.29	1.135
K5-2"	Nozzle	115.00	101.29	1.135
K7-2"	Nozzle	115.00	101.29	1.135
K2-2"	Nozzle	115.00	101.29	1.135
K4-2"	Nozzle	115.00	101.29	1.135
K6-2"	Nozzle	115.00	101.29	1.135
K8-2"	Nozzle	115.00	101.29	1.135
K10-2"	Nozzle	115.00	101.29	1.135
N1-30"	Nozzle	138.00	138.00	1.000
N1-30"	Pad	138.00	138.00	1.000
N2-30"	Nozzle	138.00	138.00	1.000
N2-30"	Pad	138.00	138.00	1.000
M1-24"	Nozzle	138.00	138.00	1.000
M1-24"	Pad	138.00	138.00	1.000
K9-2"	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Bottom Clad Dis	138.00	138.00	1.000
Clad shell 1	138.00	138.00	1.000
Unclad Shell	138.00	138.00	1.000
Unclad Topdish	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):



Description	Ambient	Operating	Ratio
D1-2"	7.93	153.00	0.052
K1-2"	2.86	153.00	0.019
K3-2"	2.86	153.00	0.019
K5-2"	2.86	153.00	0.019
K7-2"	2.86	153.00	0.019
K2-2"	2.86	153.00	0.019
K4-2"	2.86	153.00	0.019
K6-2"	2.86	153.00	0.019
K8-2"	2.86	153.00	0.019
K10-2"	2.86	153.00	0.019
N1-30"	27.85	234.00	0.119
N2-30"	27.85	234.00	0.119
M1-24"	40.35	234.00	0.172
K9-2"	2.84	225.00	0.013

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	2700	...	...	No Calc	No Calc
20	30	2700	...	...	No Calc	No Calc
30	40	No Calc	3842	17	0.00061455	61.4414
40	50	4434.33	3842	17	0.00033986	33.9784
50	60	4434.33	3842	17	0.00033986	33.9784
60	70	No Calc	3842	17	0.00061455	61.4414

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	18	No Calc	...	3.02054
40	50	18	No Calc	...	2.00451
50	60	18	No Calc	...	2.00451
60	70	18	No Calc	...	3.02054

Minimum

2

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	2700	No Calc	No Calc	No Calc
20	30	2700	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	4434.33	No Calc	No Calc	No Calc
50	60	4434.33	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	3260.01	...	3102.15	...	228.2
20	30	931.342	...	838.435	...	65.1939
30	40	3092.82	7787050	2952.24	7799033	216.498
40	50	1675.9	11379032	1583.22	11390993	117.313
50	60	4524.95	30723384	4274.69	30755680	316.747
60	70	3092.82	7787050	2952.24	7799033	216.498
Total		16577	57676516.00	15702	57744736.00	1160

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
30	Plat	4834.12	...	...	0.24851E-04	PLAT:[1 OF 1]
30	Liqd	7676.61	...	-475.75	...	Liquid: 10
30	Lini	430.277	...	-450.75	...	clad
30	Nozl	6.76348	...	-1067.51	...	D1-2"
40	Liqd	6740.54	...	300	...	Liquid: 20
40	Lini	287.069	...	500	...	clad
40	Nozl	11.5994	2006.19	800	...	K1-2"
40	Nozl	11.5994	1975.71	800	-348.371	K3-2"
40	Nozl	11.5994	1975.71	800	348.371	K5-2"
40	Nozl	11.5994	-2006.19	800	...	K7-2"
40	Nozl	11.5994	1975.71	100	-348.371	K2-2"
40	Nozl	11.5994	2006.19	100	...	K4-2"
40	Nozl	11.5994	1975.71	100	348.371	K6-2"
40	Nozl	11.5994	1885.2	100	686.157	K8-2"
40	Nozl	11.5994	-2006.19	100	...	K10-2"
40	Wght	30	...	9.99996	...	nozzles lining
50	Lini	775.086	...	1350	...	lining
50	Nozl	470.277	2053	800	...	N1-30"
50	Nozl	470.277	-2053	1400	...	N2-30"
50	Nozl	775.306	...	550	2053	M1-24"
50	Nozl	11.1989	1885.2	1550	686.157	K9-2"
50	Wght	160	...	-0.0001463	...	Donut Ring
50	Wght	250	...	-0.0001463	...	Demister pad
50	Wght	1250	1000	1000	...	N1 Distributor
50	Wght	860	-1000	1400	...	N2-Hopper
50	Wght	50	...	550	1500	Davit
60	Plat	6741.8	...	1050	...	PLAT:[1 OF 1]
60	Lini	430.277	...	500.75	...	clad
60	Wght	480	...	...	...	Clip weight

Total Weight of Each Detail Type:

Platforms	11575.9
Liquid	14417.2
Cladding & Overlay	1922.7
Nozzles	1838.2
Weights	3080.0
Sum of the Detail Weights	32834.0 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	17738.3	17738.3	17738.3	17738.3	17738.3	17738.3
Cladding	1922.7	1922.7	1922.7	1922.7	1922.7	1922.7



Nozzles	1838.2	1838.2	1838.2	1838.2	1838.2	1838.2
Wld Weights	2830.0	2830.0	2830.0	2830.0	2830.0	2830.0
Platforms	...	...	...	11575.9	11575.9	11575.9
Empty Weights	...	...	...	250.0	250.0	...
Test Weights	...	250.0	...	...	...	...
Ope Weights	...	...	...	...	...	250.0
Ope. Liquid	...	...	...	...	...	14417.2
Test Liquid	...	57641.3	...	...	...	...
Totals	24329.2	82220.5	24329.2	36155.1	36155.1	50572.3

Field Installation Options:

Platforms installed before lifting.

* Lining installed after lifting.

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	24329.2 kg.
Shop Test Wt.	- Fabricated weight + water (full)	82220.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	24329.2 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	36155.1 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	36155.1 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	50572.3 kg.
Field Test Wt.	- Empty weight + test medium (full)	93556.5 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	18066.1 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	46508.3 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	18426.4 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	204923	203641
20	30	120480	119852
30	40	166721	163001
40	50	120700	119569
50	60	325890	322836
60	70	166721	163001

Total	1105435.8	1091899.6 cm ²
Total	1189.9	1175.3 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	3488.21	3488.21	3488.21	...	3488.21
20	30	996.536	996.536	996.536	...	996.536
30	40	3316.09	16257.1	16362.8	0.00012013	8580.48
40	50	1927.61	8955.22	13586.8	7.95918	2214.68
50	60	9138.75	9913.84	40618.5	1674.42	9913.84
60	70	3789.32	10961.4	18743.7	...	10961.4

Cumulative Vessel Weight



From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	36155.1	50572.3	93796.5
20	30	32666.9	47084.1	90308.3
30	40	31670.4	46087.6	89311.7
40	50	23089.9	29830.5	72948.9
50	60	20875.2	20875.2	59362.2
60	70	10961.4	10961.4	18743.7

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	1682.38	1682.38	1682.38
20	30	1682.38	1682.38	1682.38
30	40	1682.38	1682.38	1682.38
40	50	1682.38	1682.38	1682.38
50	60	1674.42	1674.42	1674.42
60	70	...	...	...

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
D1-2"	73.86	82.30	82	600	GR 2.3	53.407	57.387	73.862	57.387
K1-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K3-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K5-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K7-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K2-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K4-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K6-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K8-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
K10-2"	36.98	41.10	82	300	GR 2.3	...	...	...	...
N1-30"	18.20	19.60	82	150	GR 1.1	11.475	17.652	18.204	15.832
N2-30"	18.20	19.60	82	150	GR 1.1	11.475	17.652	18.204	15.832
M1-24"	47.82	51.00	82	300	GR 1.1	...	...	...	...
K9-2"	47.82	51.00	82	300	GR 1.1	...	...	...	...
Min Rating	17.652	19.600, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
D1-2"	77.851	0.50	50	0	6.000	57.387	Pass
N1-30"	790.575	0.60	11353	0	6.000	17.652	Pass
N2-30"	790.575	0.60	11353	0	6.000	17.652	Pass

Evaluating Flange on Nozzle [D1-2"]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$
 $= 16(50) + 4 \cdot \text{abs}(0)77.851 \leq \pi \cdot 77.851^3 \cdot [(73.862 - 6.000) + .500 \cdot 73.862]$
 $= 0.80732E+03 \leq 0.15841E+04$ Check Passes

Maximum Pressure for Flange on Nozzle D1-2" [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(73.862, 0.500 \cdot 73.862 + 73.862 - 807.3 / (\pi \cdot 77.851^3))$
 $= \min(73.862, 57.387)$
 $= 57.387$ bars

Evaluating Flange on Nozzle [N1-30"]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$
 $= 16(11353) + 4 \cdot \text{abs}(0)790.575 \leq \pi \cdot 790.575^3 \cdot [(18.204 - 6.000) + .600 \cdot 18.204]$
 $= 0.18165E+06 \leq 0.36610E+06$ Check Passes

Maximum Pressure for Flange on Nozzle N1-30" [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(18.204, 0.600 \cdot 18.204 + 18.204 - 181646.3 / (\pi \cdot 790.575^3))$
 $= \min(18.204, 17.652)$
 $= 17.652$ bars

Evaluating Flange on Nozzle [N2-30"]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR]$



$$= 16(11353) + 4 \cdot \text{abs}(0) 790.575 \leq \pi \cdot 790.575^3 [(18.204 - 6.000) + 0.600 \cdot 18.204]$$

$$= 0.18165\text{E}+06 \leq 0.36610\text{E}+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2-30" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(18.204, 0.600 \cdot 18.204 + 18.204 - 181646.3 / (\pi \cdot 790.575^3))$$

$$= \min(18.204, 17.652)$$

$$= 17.652 \text{ bars}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 21.4235 Hz.

The Natural Frequency for the Vessel (Ope...) is 19.9598 Hz.

The Natural Frequency for the Vessel (Filled) is 13.2861 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm.
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.942) / (1 + 1.7 \cdot 3.4 \cdot 0.228)))$
 $= \min(0.85, 0.894)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	19.960 Hz
Natural Frequency of Vessel (Empty)	21.423 Hz
Natural Frequency of Vessel (Test)	13.286 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.517
Structure Height to Diameter ratio	2.015
Height to top of Structure	7469.500 mm.

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Zg = 274320.0$ mm.

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 850.0 + 300.0 = 1150.0$ mm.
 $= 3.773$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (3.773 ft.) < 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1



As there is No Hill Present: [Kzt]:

$K1 = 0, K2 = 0, K3 = 0$

Topographical Factor [Kzt]:

$= (1 + K1 \cdot K2 \cdot K3)^2$

$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$

$= 1.0$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$

$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$

$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$

Force on the first element [F]:

$= qz \cdot G \cdot Cf \cdot \text{WindArea}$

$= 16.0 \cdot 0.85 \cdot 0.517 \cdot 89.871$

$= 631.8 \text{ lbs. } [286.6] \text{ Kg}$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m2
Skirt2	1150.0	0.000	0.000	0.000	0.849	1.000	78.120
Skirt1	2500.0	0.000	0.000	0.000	0.849	1.000	78.120
Bottom Clad Dis	3025.0	0.000	0.000	0.000	0.849	1.000	78.120
Clad shell 1	3550.0	0.000	0.000	0.000	0.849	1.000	78.120
Unclad Shell	5400.0	0.000	0.000	0.000	0.879	1.000	78.120
Unclad Topdish	7184.6	0.000	0.000	0.000	0.934	1.000	78.120

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

Cf - Correction factor for natural frequency
D - Average outer insulated diameter of vessel, mm.
Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
Dr - Average outer insulated of top half of vessel, mm.
f - Natural frequency of vibration (Hertz)
f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
L - Total height of structure mm.
Lc - Total length of conical section(s) of vessel, mm.
tb - Uncorroded plate thickness at bottom of vessel, mm.
V30 - Design Wind Speed provided by user, Km/hr
Vc - Critical wind velocity, Km/hr
Vw - Maximum wind speed at top of structure, Km/hr
W - Total corroded weight of structure, Kg
Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. Kg
Z - Maximum amplitude of vibration at top of vessel, mm.
Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
Vp - Vib. Chance, <= 0.320E-06 (High); 0.320E-06 < 0.400E-06 (Probable)
P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $0.0/6500.0 = 0.0$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[12.6/21.33^2] \cdot 10^4 = 276.998$ [Geometry Violation]

Compute the vibration possibility. If $Vp > 0.400E-06$ no chance. [Vp]:

$= W / (L \cdot Dr^2)$

$= 49636 / (6500.0 \cdot 3842.001^2)$

$= 0.51733E-06$

Since Vp is > 0.400E-06 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf	Cf
PLAT:[1 OF 1]	30000.00	3000.00	78.12	234.36	1.00
PLAT:[1 OF 1]	79507.20	7800.00	78.12	621.11	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
nozzles linin	0.00	3060.00	78.12	0.00
Donut Ring	0.00	4050.00	78.12	0.00
Demister pad	0.00	4050.00	78.12	0.00
N1 Distributor	0.00	5050.00	78.12	0.00
N2-Hopper	0.00	5450.00	78.12	0.00
Davit	0.00	4600.00	78.12	0.00
Clip weight	0.00	6750.00	78.12	0.00

The Natural Frequency for the Vessel (Ope...) is 19.9598 Hz.

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm^2	Wind Pressure Kgs/m^2	Element Wind Load Kgf
10	20	1150	4911.36	83493.1	78.12	315.242
20	30	2500	4908.8	49088	78.12	185.34
30	40	3025	4917.76	2458.88	78.12	264.95
40	50	3550	4917.76	49177.6	78.12	185.678
50	60	5400	4917.76	132780	78.12	501.331
60	70	7184.57	4917.76	39904.9	78.12	892.414

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.050
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot hn^x$ where $Ct = 0.020$, $x = 0.75$ and hn = Structural Height (ft.)
 $= 0.020(24.44720.75)$
 $= 0.220$ seconds

The Coefficient Cu from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/19.96)$
 $= 0.050$

Check the Value of T which is the smaller of $Cu \cdot Ta$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.22, 0.05) \text{ per 12.8.2}$
 $= 0.050$

Loads on Building/Structure Supported Equipment:

$$\begin{aligned}
 &\text{Seismic Force per equation 13.3-1, [Fp]:} \\
 &= 0.4 \cdot Ap \cdot Sds \cdot W(1 + 2 \cdot (z/h)) / (R / Ie) \\
 &= 0.4 \cdot 2.5 \cdot 0.157 \cdot 50572(1 + 2 \cdot 0.05) / (1.25/1.25) \\
 &= 8722.711 \text{ Kgf}
 \end{aligned}$$

Maximum value of Fp per equation 13.3-2:
 $= 1.6 \cdot Sds \cdot I \cdot W$
 $= 1.6 \cdot 0.157 \cdot 1.25 \cdot 50572 = 15859.47 \text{ Kgf}$

Minimum value of Fp per equation 13.3-3:
 $= 0.3 \cdot 0.16 \cdot 1.25 \cdot 50572$
 $= 2973.652 \text{ Kgf}$

Total Base Shear $V = Fp$, [V]:
 $= 8722.711 \text{ Kgf}$

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.157 \cdot 50572$
 $= 1585.9 \text{ Kgf}$



Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 19.9598 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf	Element Emp Load Kgf	Element Vertical Load Kgf
10	20	850	3488.21	129.25	126.488	1585.95
20	30	2200	996.536	95.5705	93.5282	1476.56
30	40	2725	16257.1	1931.16	385.496	1445.31
40	50	3250	8955.22	1268.73	267.258	935.483
50	60	5100	9913.84	2204.05	1988.31	654.648
60	70	6475	10961.4	3093.96	1046.71	343.749

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm.	Cumulative Wind Shear Kgf	Earthquake Shear Kgf	Wind Bending Kg-m.	Earthquake Bending Kg-m.
10	20	850	2344.96	8722.71	10702	42247.9
20	30	2200	2029.71	8593.46	6983.5	27528.9
30	40	2725	1844.37	8497.89	5046.42	18983.1
40	50	3250	1579.42	6566.73	4960.82	18606.4
50	60	5100	1393.75	5298	3474.2	12674
60	70	6475	892.414	3093.96	387.828	1344.58

Note:
 The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear Kg	Centroid Deflection mm.	Elem. End Deflection mm.	Elem. Ang. Rotation
10	20	2344.96	0.00070675	0.0026495	0.29120E-05
20	30	2029.71	0.0043014	0.0063065	0.43259E-05
30	40	1844.37	0.0064149	0.0065236	0.43589E-05
40	50	1579.42	0.0087805	0.011176	0.49125E-05
50	60	1393.75	0.018153	0.02556	0.55593E-05
60	70	892.414	0.025699	0.025838	0.55594E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed Km/hr	2nd Crit. Wind Speed Km/hr
10	20	1759.78	10998.6
20	30	1758.86	10992.9
30	40	1762.07	11013
40	50	1762.07	11013
50	60	1762.07	11013
60	70	1762.07	11013

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 32.500 Actual deflection : 0.026 mm.

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Longitudinal Stress Constants:

From	To	Metal Area New cm ²	Metal Area Corroded cm ²	Section Modulus New mm. ³	Section Modulus Corroded mm. ³
10	20	1442	1322.19	137460944	126105312
20	30	1201.65	1081.78	114609376	103230112
30	40	2162.43	2042.84	205764080	194485536
40	50	2162.43	2042.84	205764080	194485536
50	60	2162.43	2042.84	205764080	194485536
60	70	2162.43	2042.84	205764080	194485536

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Longitudinal Allowable Stresses:

From	To	Tensile N./mm ²	Hydrotest Tensile N./mm ²	Compressive N./mm ²	Hydrotest Compressive N./mm ²
10	20	95.7583	188.997	-85.9853	-85.9853
20	30	82.7985	140.397	-70.3883	-70.3883
30	40	140.76	238.68	-106.011	-106.011
40	50	140.76	238.68	-106.011	-106.011
50	60	140.76	238.68	-106.011	-106.011
60	70	140.76	238.68	-106.011	-106.011

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure N./mm ²	Longitudinal Stress External Pressure N./mm ²	Longitudinal Stress Hydrotest Pressure N./mm ²
10	20	...	...	...
20	30	...	...	...
30	40	55.8032	...	72.5442
40	50	55.8032	...	72.5442
50	60	55.8032	...	72.5442
60	70	55.8032	...	72.5442

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty N./mm ²	Wght. Str. Operating N./mm ²	Wght. Str. Hydrotest N./mm ²	Wght. Str. Emp. Mom. N./mm ²	Wght. Str. Opr. Mom. N./mm ²
10	20	-2.68167	-3.751	-6.95699	0.13083	0.13083
20	30	-2.96141	-4.26838	-8.18685	0.15982	0.15982
30	40	-1.52036	-1.52036	-1.52036	0.084831	0.084831
40	50	-1.10845	-1.10845	-1.10845	0.084831	0.084831
50	60	-1.00213	-1.00213	-1.00213	0.08443	0.08443
60	70	-0.52621	-0.52621	-0.52621	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. N./mm ²	Bend. Str. Oper. Wind N./mm ²	Bend. Str. Oper. Equ. N./mm ²	Bend. Str. Hyd. Wind N./mm ²	Bend. Str. Hyd. Equ. N./mm ²
10	20	0.13083	0.83225	3.28543	0.41612	...
20	30	0.15982	0.66342	2.61519	0.33171	...
30	40	0.084831	0.25446	0.95719	0.12723	...
40	50	0.084831	0.25014	0.9382	0.12507	...
50	60	0.08443	0.17518	0.63906	0.087591	...
60	70	...	0.019556	0.067798	0.0097778	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case N./mm ²	Vortex Shedding Empty Case N./mm ²	Vortex Shedding Test Case N./mm ²	Earthquake Empty Case N./mm ²
10	20	...	...	...	1.52259
20	30	...	...	...	1.2391
30	40	...	...	...	0.46938
40	50	...	...	...	0.46057
50	60	...	...	...	0.30079
60	70	...	...	...	0.022937

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind	Longitudinal Stress Y Forces Seismic
------	----	--------------------------------------	---

		N./mm ²	N./mm ²
10	20	...	0.11763
20	30	...	0.13386
30	40	...	0.069383
40	50	...	0.044909
50	60	...	0.031427
60	70	...	0.016502

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded N./mm ²	Earthquake For/Mom Corroded N./mm ²	Wind For/Mom No Corrosion N./mm ²	Earthquake For/Mom No Corrosion N./mm ²
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress N./mm ²
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...

**Stress Combination Load Cases for Vertical Vessels:****Load Case Definition Key**

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.64	85.99		0.0424
20		82.80	-3.78	70.39		0.0538
30		140.76	-1.86	106.01		0.0175
40		140.76	-1.44	106.01		0.0136
50		140.76	-1.26	106.01		0.0119
60		140.76	-0.55	106.01		0.0051

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.22	85.99		0.0490
20		82.80	-4.23	70.39		0.0600
30		140.76	-2.01	106.01		0.0189
40		140.76	-1.61	106.01		0.0152
50		140.76	-1.36	106.01		0.0128
60		140.76	-0.53	106.01		0.0050

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.71	85.99		0.0548



20	82.80	-5.09	70.39	0.0723
30	140.76	-1.86	106.01	0.0175
40	140.76	-1.44	106.01	0.0136
50	140.76	-1.26	106.01	0.0119
60	140.76	-0.55	106.01	0.0051

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-7.05	85.99		0.0820
20		82.80	-6.91	70.39		0.0982
30		140.76	-2.49	106.01		0.0235
40		140.76	-2.09	106.01		0.0197
50		140.76	-1.69	106.01		0.0160
60		140.76	-0.58	106.01		0.0054

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-7.50	85.99		0.0873
20		140.40	-8.68	70.39		0.1233
30		238.68	-1.73	106.01		0.0163
40		238.68	-1.32	106.01		0.0124
50		238.68	-1.17	106.01		0.0111
60		238.68	-0.54	106.01		0.0051

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-7.09	85.99		0.0824
20		140.40	-8.35	70.39		0.1186
30		238.68	-1.61	106.01		0.0151
40		238.68	-1.19	106.01		0.0113
50		238.68	-1.09	106.01		0.0102
60		238.68	-0.53	106.01		0.0050

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.71	85.99		0.0548
20		82.80	-5.09	70.39		0.0723
30	54.62	140.76		106.01	0.3881	
40	55.03	140.76		106.01	0.3909	
50	55.06	140.76		106.01	0.3912	
60	55.30	140.76		106.01	0.3928	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-7.05	85.99		0.0820
20		82.80	-6.91	70.39		0.0982
30	55.39	140.76		106.01	0.3935	
40	55.76	140.76		106.01	0.3962	
50	55.56	140.76		106.01	0.3947	
60	55.36	140.76		106.01	0.3933	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-4.71	85.99		0.0548
20		82.80	-5.09	70.39		0.0723
30		140.76	-1.86	106.01		0.0175
40		140.76	-1.44	106.01		0.0136



50	140.76	-1.26	106.01	0.0119
60	140.76	-0.55	106.01	0.0051

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-7.05	85.99		0.0820
20		82.80	-6.91	70.39		0.0982
30		140.76	-2.49	106.01		0.0235
40		140.76	-2.09	106.01		0.0197
50		140.76	-1.69	106.01		0.0160
60		140.76	-0.58	106.01		0.0054

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-7.50	85.99		0.0873
20		140.40	-8.68	70.39		0.1233
30	71.24	238.68		106.01	0.2985	
40	71.65	238.68		106.01	0.3002	
50	71.71	238.68		106.01	0.3005	
60	72.03	238.68		106.01	0.3018	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		189.00	-7.09	85.99		0.0824
20		140.40	-8.35	70.39		0.1186
30	71.11	238.68		106.01	0.2979	
40	71.52	238.68		106.01	0.2997	
50	71.63	238.68		106.01	0.3001	
60	72.02	238.68		106.01	0.3017	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-2.81	85.99		0.0327
20		82.80	-3.12	70.39		0.0443
30	54.37	140.76		106.01	0.3862	
40	54.78	140.76		106.01	0.3892	
50	54.89	140.76		106.01	0.3899	
60	55.28	140.76		106.01	0.3927	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.44	85.99		0.0400
20		82.80	-3.84	70.39		0.0546
30	54.37	140.76		106.01	0.3862	
40	54.76	140.76		106.01	0.3890	
50	54.86	140.76		106.01	0.3897	
60	55.31	140.76		106.01	0.3929	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.88	85.99		0.0451
20		82.80	-4.43	70.39		0.0629
30	54.37	140.76		106.01	0.3862	
40	54.78	140.76		106.01	0.3892	
50	54.89	140.76		106.01	0.3899	
60	55.28	140.76		106.01	0.3927	



Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-2.81	85.99		0.0327
20		82.80	-3.12	70.39		0.0443
30	54.37	140.76		106.01	0.3862	
40	54.78	140.76		106.01	0.3892	
50	54.89	140.76		106.01	0.3899	
60	55.28	140.76		106.01	0.3927	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.88	85.99		0.0451
20		82.80	-4.43	70.39		0.0629
30		140.76	-1.61	106.01		0.0151
40		140.76	-1.19	106.01		0.0113
50		140.76	-1.09	106.01		0.0102
60		140.76	-0.53	106.01		0.0050

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-3.76	71.65		0.0525
20		69.00	-4.29	58.66		0.0732
30	54.44	117.30		88.34	0.4641	
40	54.82	117.30		88.34	0.4674	
50	54.92	117.30		88.34	0.4682	
60	55.29	117.30		88.34	0.4714	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		79.80	-3.76	71.65		0.0525
20		69.00	-4.29	58.66		0.0732
30		117.30	-1.54	88.34		0.0174
40		117.30	-1.15	88.34		0.0130
50		117.30	-1.06	88.34		0.0119
60		117.30	-0.51	88.34		0.0058

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		95.76	-3.88	85.99		0.0451
20		82.80	-4.43	70.39		0.0629
30	54.37	140.76		106.01	0.3862	
40	54.78	140.76		106.01	0.3892	
50	54.89	140.76		106.01	0.3899	
60	55.28	140.76		106.01	0.3927	

Absolute Maximum of the all of the Stress Ratio's

0.4714

Governing Element: Unclad Topdish

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt2	...	-3.6	0.1	...	...	+ Tens
Skirt2	...	-3.6	-0.1	...	-3.8	- Comp
Skirt1	...	-4.1	0.2	...	...	+ Tens
Skirt1	...	-4.1	-0.2	...	-4.3	- Comp



Bottom Clad Dis	55.8	-1.5	0.1	...	54.4	+ Tens
Bottom Clad Dis	55.8	-1.5	-0.1	...	...	- Comp
Clad shell 1	55.8	-1.1	0.1	...	54.8	+ Tens
Clad shell 1	55.8	-1.1	-0.1	...	...	- Comp
Unclad Shell	55.8	-1.0	0.1	...	54.9	+ Tens
Unclad Shell	55.8	-1.0	-0.1	...	...	- Comp
Unclad Topdish	55.8	-0.5	...	...	55.3	+ Tens
Unclad Topdish	55.8	-0.5	...	...	...	- Comp

**Skirt Support Analysis Per EN 13445-3 Section 16.12:**

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt1

Total vessel (empty) weight	Fg	31963.799 Kg
Head Weight below section 2-2	DeltaFg	8580.484 Kg
Weight of vessel contents (liquid)	Ff	14417.154 Kg
Vessel Gross Weight	Fg + Ff	46380.953 Kg

Skirt Mean Diameter	Dz	3826.000 mm.
Skirt Analysis Thickness (Corroded)	ez	9.000 mm.
Skirt Height	H	1000.000 mm.

Additional Axial Force at section 1-1	F1	0.000 Kg
Moment at section 1-1	M1	18983.061 Kg-m.
Applied Moment - Bottom of Skirt	Mb	27528.906 Kg-m.
Weight at Bottom of Skirt	W	47084.098 Kg

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	3825.000 mm.
Head Corroded Thickness	eb	17.000 mm.

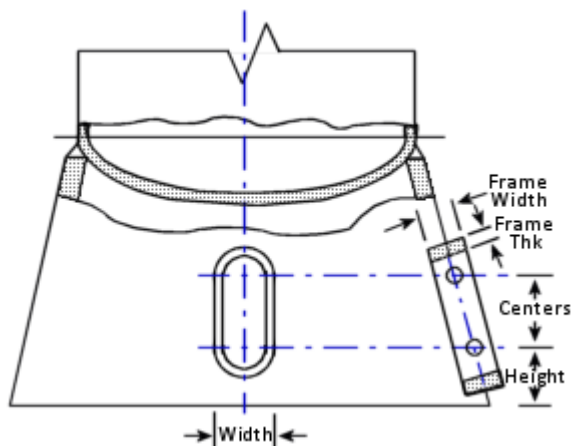
Per EN 13445 Section 7.5.4 [R, r]:

$$h_i = 952.5 \text{ mm.}$$

$$K = D_i / (2 * h_i) = 3808.0 / (2 * 952.5) = 2.0$$

$$R = D_i (0.44 * K + 0.02) = 3808.0 * (0.44 * 2.0 + 0.02) = 3425.4409 \text{ mm.}$$

$$r = D_i ((0.5 / K) - 0.08) = 3808.0 * ((0.5 / 2.0) - 0.08) = 647.86 \text{ mm.}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	19794152.000 cm ⁴
Moment of Inertia	Iyy =	19794152.000 cm ⁴
Section Modulus	W =	103471864.000 mm. ³
Section Metal Area	A =	1081.776 cm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})}$$

$$= \sqrt{(\min(19794152, 19794152) / 1081.776)}$$

$$= 1352.695 \text{ mm.}$$



Skirt Slenderness Ratio [rg/L]:

$$= r_g / L$$

$$= 1352.695 / 1000.0$$

$$= 1.353$$

Correct the Moment [M4]:

top of skirt is the critical end
Controlling Moment [M4] = 18983.061 Kg-m.

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -18983.061 / 41389E+09 - 47084.098 / 1081.77612$$

$$= -4.718 \text{ N./mm}^2$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +18983.061 / 41389E+09 - 47084.098 / 1081.77612$$

$$= -3.819 \text{ N./mm}^2$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= 0.0 - 31963.799 - 14417.154 + 4 \cdot 18983.061 / 3826.0$$

$$= -26535.0 \text{ Kgf}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= 0.0 - 31963.799 - 14417.154 - 4 \cdot 18983.061 / 3826.0$$

$$= -66226.9 \text{ Kgf}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-26534.969 + 8580.484 + 14417.154) / (\pi \cdot 3825.0 \cdot 17.0) + 10.0 \cdot 3825.0 / (4 \cdot 17.0)$$

$$= 56.08 \text{ N./mm}^2$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-66226.938 + 8580.484 + 14417.154) / (\pi \cdot 3825.0 \cdot 17.0) + 10.0 \cdot 3825.0 / (4 \cdot 17.0)$$

$$= 54.18 \text{ N./mm}^2$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (14417.154 + 8580.484) / (\pi \cdot 3825.0 \cdot 17.0) + 10.0 \cdot 3825.0 / (4 \cdot 17.0)$$

$$= 57.36 \text{ N./mm}^2$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= -26534.969 / (\pi \cdot 3826.0 \cdot 9.0)$$

$$= -2.41 \text{ N./mm}^2$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -66226.938 / (\pi \cdot 3826.0 \cdot 9.0)$$

$$= -6.00 \text{ N./mm}^2$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$



$$= 1 - 0.5(3825.0 + 17.0 - 3826.0 + 9.0) / (647.86 + 17.0)$$

$$= 0.9812$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.981) = 11.128 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(17.0^2 + 9.0^2 + 2 \cdot 17.0 \cdot 9.0 \cdot 0.981)}$$

$$= 12.945 \text{ mm.}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 12.94 \cdot -26535.0 = -343.49 \text{ Kg-m.}$$

$$M_q = a \cdot F_{zq} = 12.94 \cdot -66226.9 = -857.30 \text{ Kg-m.}$$

Correction factor [C]:

$$= 0.63 - 0.057(eb/ez)^2$$

$$= 0.63 - 0.057(17.0/9.0)^2$$

$$= 0.427$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.427 \cdot 6 \cdot -343.49 / (\pi \cdot 3825.0 \cdot 17.0^2)$$

$$= -2.48 \text{ N./mm}^2$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2)$$

$$= 0.427 \cdot 6 \cdot -857.296 / (\pi \cdot 3825.0 \cdot 17.0^2)$$

$$= -6.20 \text{ N./mm}^2$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.427 \cdot 6 \cdot -343.49 / (\pi \cdot 3826.0 \cdot 9.0^2)$$

$$= -8.86 \text{ N./mm}^2$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2)$$

$$= 0.427 \cdot 6 \cdot -857.296 / (\pi \cdot 3826.0 \cdot 9.0^2)$$

$$= -22.10 \text{ N./mm}^2$$

Geometry variables, 16.12-32 to 36:

$$eb/D_b = 0.669/150.591 = 0.0044$$

$$y = 125 \cdot eb/D_b = 125 \cdot 17.0/3825.0 = 0.5556$$

As this an elliptical head, it will be treated a a Korbbogen type head

Geometric value [α]:

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994$$

$$= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.556 + 3.994 = 4.1385$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= (P + Ph) \cdot D_b \cdot ((y/\gamma A) \cdot \alpha - 1) / (4 \cdot eb)$$

$$= (10.0 + 0.0) \cdot 3825.0 \cdot ((11.13/40.0) \cdot 4.139 - 1) / (4 \cdot 17.0)$$

$$= 8.51 \text{ N./mm}^2$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 56.083 - 2.483 + 8.513$$

$$= 67.08 \text{ N./mm}^2$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 56.083 + 2.483 - 8.513$$

$$= 45.09 \text{ N./mm}^2$$



Stress [σ_{1qitot}]:

$$\begin{aligned}
 &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\
 &= 54.178 - -6.197 + 8.513 \\
 &= 68.89 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned}
 &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\
 &= 54.178 + -6.197 - 8.513 \\
 &= 39.47 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{2pitot}]:

$$\begin{aligned}
 &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\
 &= 57.357 + -2.483 + 8.513 \\
 &= 63.39 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 57.357 - -2.483 - 8.513 \\
 &= 51.33 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{2qitot}]:

$$\begin{aligned}
 &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\
 &= 57.357 + -6.197 + 8.513 \\
 &= 59.67 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{2qotot}]:

$$\begin{aligned}
 &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\
 &= 57.357 - -6.197 - 8.513 \\
 &= 55.04 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{3pitot}]:

$$\begin{aligned}
 &= \sigma_{3pm} - \sigma_{3pb} \\
 &= -2.406 - -8.856 \\
 &= 6.45 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{3potot}]:

$$\begin{aligned}
 &= \sigma_{3pm} + \sigma_{3pb} \\
 &= -2.406 + -8.856 \\
 &= -11.26 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{3qitot}]:

$$\begin{aligned}
 &= \sigma_{3qm} - \sigma_{3qb} \\
 &= -6.004 - -22.104 \\
 &= 16.10 \text{ N./mm}^2
 \end{aligned}$$

Stress [σ_{3qotot}]:

$$\begin{aligned}
 &= \sigma_{3qm} + \sigma_{3qb} \\
 &= -6.004 + -22.104 \\
 &= -28.11 \text{ N./mm}^2
 \end{aligned}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$\begin{aligned}
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\
 &= 138.0(3 - (1/1.5)) \cdot (56.083/137.997)^2 \\
 &= 398.80 \text{ N./mm}^2
 \end{aligned}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$\begin{aligned}
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\
 &= 138.0(3 - (1/1.5)) \cdot (54.178/137.997)^2 \\
 &= 399.82 \text{ N./mm}^2
 \end{aligned}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$\begin{aligned}
 &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\
 &= 138.0(3 - (1/1.5)) \cdot (57.357/137.997)^2 \\
 &= 398.11 \text{ N./mm}^2
 \end{aligned}$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:



$$= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (57.357/137.997)^2)$$

$$= 398.11 \text{ N./mm}^2$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (-2.406/137.997)^2)$$

$$= 413.97 \text{ N./mm}^2$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138.0(3 - (1/1.5) \cdot (-6.004/137.997)^2)$$

$$= 413.83 \text{ N./mm}^2$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (N./mm²)

Stress	Actual	Allowed	Result	Equation
Stot1pi	67.08	398.80	Pass	46
Stot1po	45.09	398.80	Pass	47
Stot1qi	68.89	399.82	Pass	48
Stot1qo	39.47	399.82	Pass	49
Stot2pi	63.39	398.11	Pass	50
Stot2po	51.33	398.11	Pass	51
Stot2qi	59.67	398.11	Pass	52
Stot2qo	55.04	398.11	Pass	53
Stot3pi	6.45	413.97	Pass	54
Stot3po	-11.26	413.97	Pass	55
Stot3qi	16.10	413.83	Pass	56
Stot3qo	-28.11	413.83	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
 Lining is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	5495.513 mm.
Center of Gravity of the Liquid	2610.318 mm.
Center of Gravity of the Lining	4600.000 mm.
Center of Gravity of the Nozzles	4513.015 mm.
Center of Gravity of the Added Weights (Operating)	4966.817 mm.
Center of Gravity of the Added Weights (Empty)	4966.817 mm.
Center of Gravity of Bare Shell New and Cold	3727.738 mm.
Center of Gravity of Bare Shell Corroded	3731.037 mm.
Vessel CG in the Operating Condition	3956.322 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	3959.895 mm.
Vessel CG in the Test Condition	4419.391 mm.

Rigging Analysis Results:

• Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	6500.00 mm.
Total vessel weight (No Liquid)	36155.14 Kgf
Impact weight multiplication factor	1.50
Design lifting weight, DWT = Imp * Twt	54232.71 Kgf
Elevation of the Tailing Lug (bottom)	150.00 mm.
Elevation of the Lifting Lug (top)	6160.00 mm.
Design Reaction force at the tailing lug	19853.20 Kgf
Design Reaction force at the lifting lug	34379.52 Kgf
CG distance during field lifting operation	3959.89 mm.
CG distance from tailing lug	3809.89 mm.
CG distance from the nearer lifting lug/trunnion	2200.11 mm.

Compressive Allowable Increase Factor 1.00

Critical Values: (N./mm², mm.)

	Max Stress	Elevation	Allowables
Bending	2.27	2500.00	65.17 (UG-23)
Shear	1.14	5910.00	96.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm.	Bending Moment Kg-m.	Bending Stress N./mm ²	Shear Force Kgf	Shear Stress N./mm ²
0.00	0.0	0.0	14740.4	1.0
1700.00	19710.9	1.7	423.2	0.0
2700.00	24920.8	1.2	3020.5	0.1
2750.00	25011.7	1.2	3020.5	0.1
3750.00	24521.7	1.2	7634.6	0.3
6450.00	3654.0	0.2	1209.4	0.1

Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0348 at elevation 2500.0 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Center of Gravity Calculation: Step: 23 2:27pm Sep 16,2026

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HEXAGON

--- Plot data successfully generated ...----

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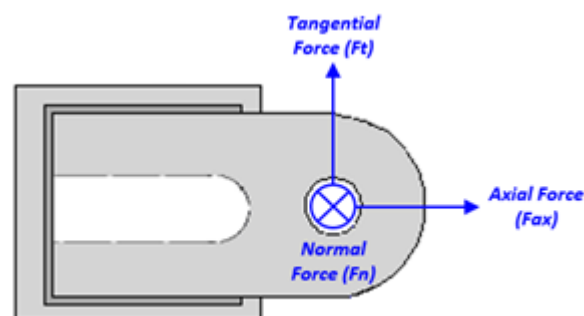
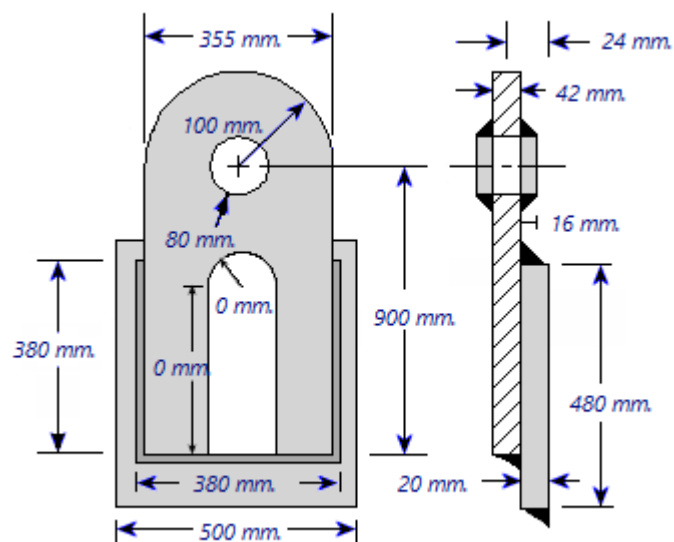
Lifting Lug Calculations:

Input Values:

Lifting Lug Material		SA-516 70
Lifting Lug Yield Stress	Yield	262.00 N./mm ²
Width of Lifting Lug	w	355.0000 mm.
Thickness of Lifting Lug	t	42.0000 mm.
Diameter of Hole in Lifting Lug	dh	80.0000 mm.
Radius of Semi-Circular Arc of Lifting Lug	r	100.0000 mm.
Height of Lug from Bottom to Center of Hole	h	900.0000 mm.
Offset from Vessel OD to Center of Hole	off	24.0000 mm.
Lug Fillet Weld Size	tw	20.0000 mm.
Length of weld along side of Lifting Lug	wl	380.0000 mm.
Length of Weld along Bottom of Lifting Lug	wb	380.0000 mm.
Thickness of Collar (if any)	tc	0.0000 mm.
Diameter of Collar (if any)	dc	0.0000 mm.
Impact Factor	Impfac	1.50
Number of Lugs in Group		2
Reinforcing Pad Circumferential Width	wpb	500.00 mm.
Reinforcing Pad Longitudinal Length	wpl	480.00 mm.
Reinforcing Pad Thickness	thkp	20.00 mm.
Reinforcing Pad Fillet Weld Size	tfp	16.00 mm.

Lifting Lug Orientation to Vessel: Flat

Lift Orientation : From Horizontal to Vertical



PV Elite computes the forces and stresses from 0 to 90 degrees at 1 degree increments. The table below provides the results in 5 degree increments.

The 0 degree angle position is in the horizontal direction and the 90 degree angle position is in the vertical direction.

PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

Lifting Lug and Weld Stresses at various Angles: N./mm²

Angle	----- Weld Stresses			----- Lug Stresses		Maximum
	Primary	Secondary	Allowable	Combined	Allowable	Ratio
0	12.3	72.2	104.8	99.4	172.9	0.689
5	12.8	73.2	104.8	101.7	172.9	0.698
10	13.4	73.5	104.8	103.2	172.9	0.702
15	13.9	73.3	104.8	103.9	172.9	0.699
20	14.4	72.4	104.8	103.7	172.9	0.690
25	14.9	70.8	104.8	102.7	172.9	0.676
30	15.3	68.7	104.8	100.8	172.9	0.656
35	15.8	66.0	104.8	98.0	172.9	0.630
40	16.2	62.7	104.8	94.5	172.9	0.598
45	16.7	58.8	104.8	90.2	172.9	0.561
50	17.1	54.4	104.8	85.0	172.9	0.519
55	17.5	49.5	104.8	79.2	172.9	0.472
60	17.9	44.1	104.8	72.6	172.9	0.421
65	18.4	38.2	104.8	65.3	172.9	0.378
70	18.9	31.8	104.8	57.4	172.9	0.332
75	19.5	24.9	104.8	48.7	172.9	0.282
80	20.2	17.4	104.8	39.4	172.9	0.228
85	21.2	9.2	104.8	29.2	172.9	0.202
90	22.6	2.0	104.8	17.8	172.9	0.215

Computed Results:

Total vessel weight (no liquid)	36155.14 Kg
Design reaction force at the tailing lug	19853.20 Kg
Design reaction force at the lifting lug	34379.52 Kg

Worst case lift angle for primary weld stresses	90 deg
Tangential force at this angle	0.0 Kg
Axial force at this angle	27116.4 Kg

Worst case lift angle for secondary weld stresses	10 deg
Tangential force at this angle	17507.8 Kg
Axial force at this angle	3087.1 Kg

Worst case lift angle for lug stresses	16 deg
Tangential force at this angle	17410.3 Kg
Axial force at this angle	4992.3 Kg

Converting the weld leg dimension (tw) to the weld throat dimension.

Weld Group Inertia Calculations:

Weld group inertia about the longitudinal axis	I _{ll}	26852.211 cm ⁴
Weld group centroid distance in the long. direction	Y _{ll}	255.690 mm.
Dist. of weld group centroid from lug bottom	Y _{llb}	124.310 mm.
Weld group inertia about the circumferential axis	I _{lc}	42444.711 cm ⁴
Weld group centroid distance in the circ. direction	Y _{lc}	190.000 mm.

Note: The Impact Factor is applied to the Forces acting on the Lug.

Primary Shear Stress in the Welds due to Shear Loads [S_{sl}]:

$$\begin{aligned}
 &= \sqrt{(F_{ax}^2 + F_t^2 + F_n^2)} / ((2 \cdot w_l + w_b) \cdot t_w) \\
 &= \sqrt{(27116^2 + 0^2 + 0^2)} / ((2 \cdot 380.0 + 380.0) \cdot 14.14) \\
 &= 16.50 \text{ N./mm}^2
 \end{aligned}$$



Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (F_n(h-Y_{IIb}))Y_{II/III} + (F_{ax} \cdot \text{off} \cdot Y_{II/III}) + (F_t \cdot \text{off} \cdot Y_{Ic/IIc})$$

$$= (0(900.0-124.31))255.69/26852.211 +$$

$$(27116 \cdot 24.0 \cdot 255.69/26852.211) +$$

$$(0 \cdot 24.0 \cdot 190.0/42444.711)$$

$$= 6.08 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= S_{sII} + S_{blf}$$

$$= 16.497 + 6.077$$

$$= 22.57 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Secondary Shear Stress in the Welds due to Shear Loads:

Unit Weld Polar Section Modulus [Uwsm]:

$$= (2 \cdot w_l + w)^3 / 12 - w_l^2(w_l + w)^2 / (2 \cdot w_l + w)$$

$$= 45553568.00 \text{ mm}^3$$

Loads on Welds due to Torsional Moment:

Shear Stress in Normal Direction [Fth]:

$$= F_t \cdot (h - (w_l - \text{Cent})) \cdot (w_b/2) / U_{wsm}$$

$$= 56.26 \text{ Kg/mm.}$$

Shear Stress in Circumferential Direction [Ftv]:

$$= F_t \cdot (h - (w_l - \text{Cent})) \cdot \text{Cent} / U_{wsm}$$

$$= 74.18 \text{ Kg/mm.}$$

Primary Shear Stress [Fsv]:

$$= F_t / (2 \cdot w_l + w)$$

$$= 15.70 \text{ Kg/mm.}$$

Resultant Load on Weld Group [Fr]:

$$= \sqrt{(F_{th}^2 + (F_{tv} + F_{sv})^2)}$$

$$= \sqrt{(56.3^2 + (74.2 + 15.7)^2)}$$

$$= 106.04 \text{ Kg/mm.}$$

Resultant Secondary Weld Stress [Fws]:

$$= F_r / T_{weld}$$

$$= 106.038/14.14$$

$$= 73.54 \text{ N./mm}^2$$

Allowable Resultant Secondary Weld Stress [Psa]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(P_l^2 + F_n^2)} / S_{ha}$$

$$= \sqrt{(27116^2 + 0^2)} / 50.4$$

$$= 52.76 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 42.0 + 2 \cdot 0.0$$

$$= 42.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:



$$\begin{aligned}
 &= \sqrt{(P_l^2 + F_n^2)} / (P_{bw} \cdot d_h) \\
 &= \sqrt{(27116^2 + 0^2)} / (42.0 \cdot 80.0) \\
 &= 79.14 \text{ N./mm}^2
 \end{aligned}$$

Allowable Bearing Stress [Pba]:

$$\begin{aligned}
 &= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.} \\
 &= \min(0.75 \cdot 262 \cdot 1.0, 235.8) \\
 &= 196.50 \text{ N./mm}^2
 \end{aligned}$$

Bending Stress at the Base of the Lug [Fbs]:

$$\begin{aligned}
 &= F_n \cdot (h - w_l) / (w \cdot t^2 / 6) + F_t \cdot (h - w_l) / (w^2 \cdot t / 6) \\
 &= 0 \cdot (900.0 - 380.0) / (355.0 \cdot 42.0^2 / 6) + \\
 &\quad 17410 \cdot (900.0 - 380.0) / (355.0^2 \cdot 42.0 / 6) \\
 &= 100.64 \text{ N./mm}^2
 \end{aligned}$$

Tensile Stress at the Base of the Lug [Fa]:

$$\begin{aligned}
 &= F_{ax} / (w \cdot t) \\
 &= 4992 / (355.0 \cdot 42.0) \\
 &= 3.28 \text{ N./mm}^2
 \end{aligned}$$

Total Combined Stress at the Base of the Lug:

$$\begin{aligned}
 &= F_{bs} + F_a \\
 &= 100.6 + 3.3 \\
 &= 103.93 \text{ N./mm}^2
 \end{aligned}$$

Lug Allowable Stress for Bending and Tension:

$$\begin{aligned}
 &= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield}) \\
 &= \min(0.66 \cdot 262 \cdot 1.0, 196.5) \\
 &= 172.92 \text{ N./mm}^2
 \end{aligned}$$

Required Shackle Pin Diameter [Spd]:

$$\begin{aligned}
 &= \sqrt{((2(P_l^2 + F_n^2))^{1/2} / (\pi \cdot \text{Sta}))} \\
 &= \sqrt{(2(27116^2 + 0^2))^{1/2} / (\pi \cdot 105)} \\
 &= 40.1924 \text{ mm.}
 \end{aligned}$$

WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).

The full lug load is applied in both the circ. and long. directions simultaneously.

Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.

Input Echo, WRC107/537 Item 1, Description: Lift Lug

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Cylindrical
Internal Corrosion Allowance	Cas	0.0000 mm.
Vessel Diameter	Dv	3806.000 mm.
Vessel Thickness	Tv	18.000 mm.

Note: Lifting lug calculations are performed at ambient temperature conditions.

Attachment Type	Type	Rectangular
Parameter C11	C11	395.00 mm.
Parameter C22	C22	400.00 mm.
Thickness of Reinforcing Pad	Tpad	20.000 mm.
Pad Parameter C11P	C11p	532.000 mm.
Pad Parameter C22P	C22p	496.000 mm.
Design Internal Pressure	Dp	0.000 bars
Include Pressure Thrust		No

External Forces and Moments in WRC 107/537 Convention:

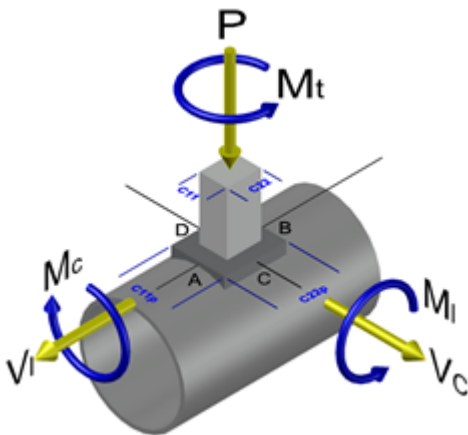
Radial Load (SUS)	P	0.0 Kg
Longitudinal Shear (SUS)	Vl	27116.4 Kg
Circumferential Shear (SUS)	Vc	27116.4 Kg
Circumferential Moment (SUS)	Mc	0.0 Kg-m.

Longitudinal Moment (SUS) Ml 0.0 Kg-m.
 Torsional Moment (SUS) Mt 19253.0 Kg-m.

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	0.0	Kgf
Circumferential Shear	VC	27116.4	Kgf
Longitudinal Shear	VL	27116.4	Kgf
Circumferential Moment	MC	0.0	Kg-m.
Longitudinal Moment	ML	0.0	Kg-m.
Torsional Moment	MT	19253.0	Kg-m.

Dimensionless Parameters used : Gamma = 50.58

Dimensionless Loads for Cylindrical Shells at Attachment Junction:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.104	4C	8.574	(A,B)
N(PHI) / (P/Rm)	0.104	3C	7.311	(C,D)
M(PHI) / (P)	0.103	2C1	0.074	(A,B)
M(PHI) / (P)	0.103	1C	0.109	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.103	3A	1.729	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.114	1A	0.089	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.104	3B	5.775	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.106	1B	0.041	(A,B,C,D)
N(x) / (P/Rm)	0.104	3C	7.330	(A,B)
N(x) / (P/Rm)	0.104	4C	8.582	(C,D)
M(x) / (P)	0.104	1C1	0.113	(A,B)
M(x) / (P)	0.104	2C	0.074	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.103	4A	2.496	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.123	2A	0.045	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.104	4B	1.640	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.109	2B	0.064	(A,B,C,D)



Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Memb. P		-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P		-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC		0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC		0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML		-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML		-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear VC		8.9	8.9	-8.9	-8.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-8.7	-8.7	8.7	8.7
Shear MT		125.8	125.8	125.8	125.8	125.8	125.8	125.8	125.8
Tot. Shear		134.6	134.6	116.9	116.9	117.0	117.0	134.5	134.5
Str. Int.		269.3	269.3	233.9	233.9	234.1	234.1	269.1	269.1

Dimensionless Parameters used : $\Gamma = 106.22$

Dimensionless Loads for Cylindrical Shells at Pad Edge:

Curves read for 1979	Beta	Figure	Value	Location
N(PHI) / (P/Rm)	0.134	4C	14.391	(A,B)
N(PHI) / (P/Rm)	0.134	3C	8.979	(C,D)
M(PHI) / (P)	0.137	2C1	0.033	(A,B)
M(PHI) / (P)	0.137	1C !	0.067	(C,D)
N(PHI) / (MC/(Rm**2 * Beta))	0.136	3A	4.159	(A,B,C,D)
M(PHI) / (MC/(Rm * Beta))	0.144	1A	0.069	(A,B,C,D)
N(PHI) / (ML/(Rm**2 * Beta))	0.133	3B	10.530	(A,B,C,D)
M(PHI) / (ML/(Rm * Beta))	0.135	1B	0.022	(A,B,C,D)
N(x) / (P/Rm)	0.137	3C	8.783	(A,B)
N(x) / (P/Rm)	0.137	4C	14.267	(C,D)
M(x) / (P)	0.135	1C1	0.065	(A,B)
M(x) / (P)	0.135	2C !	0.037	(C,D)
N(x) / (MC/(Rm**2 * Beta))	0.136	4A	8.373	(A,B,C,D)
M(x) / (MC/(Rm * Beta))	0.155	2A	0.030	(A,B,C,D)
N(x) / (ML/(Rm**2 * Beta))	0.133	4B	4.242	(A,B,C,D)
M(x) / (ML/(Rm * Beta))	0.137	2B	0.028	(A,B,C,D)

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Memb. P	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Circ. Bend. P	-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Circ. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Circ. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Circ. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Circ. Str.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Memb. P	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0	-0.0
Long. Bend. P	-0.0	0.0	-0.0	0.0	-0.0	0.0	-0.0	0.0
Long. Memb. MC	0.0	0.0	0.0	0.0	-0.0	-0.0	0.0	0.0
Long. Bend. MC	0.0	0.0	0.0	0.0	-0.0	0.0	0.0	-0.0
Long. Memb. ML	-0.0	-0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Bend. ML	-0.0	0.0	0.0	-0.0	0.0	0.0	0.0	0.0
Tot. Long. Str.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear VC	13.9	13.9	-13.9	-13.9	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-14.9	-14.9	14.9	14.9
Shear MT	159.0	159.0	159.0	159.0	159.0	159.0	159.0	159.0
Tot. Shear	172.9	172.9	145.1	145.1	144.1	144.1	173.9	173.9
Str. Int.	345.8	345.8	290.2	290.2	288.2	288.2	347.8	347.8

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		8.9	8.9	-8.9	-8.9	-8.7	-8.7	8.7	8.7
Shear Q (SUS)		125.8	125.8	125.8	125.8	125.8	125.8	125.8	125.8
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		17.7	17.7	17.7	17.7	17.5	17.5	17.5	17.5
Pm+Pl+Q (Total)		269.3	269.3	233.9	233.9	234.1	234.1	269.1	269.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	17.72	207.00	Passed
Pm+Pl+Q (TOTAL)	269.29	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (N/mm²)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Circ. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Pl (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Long. Q (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		13.9	13.9	-13.9	-13.9	-14.9	-14.9	14.9	14.9
Shear Q (SUS)		159.0	159.0	159.0	159.0	159.0	159.0	159.0	159.0
Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Pm+Pl (SUS)		27.8	27.8	27.8	27.8	29.8	29.8	29.8	29.8
Pm+Pl+Q (Total)		345.8	345.8	290.2	290.2	288.2	288.2	347.8	347.8

Vessel Stress Summation Comparison (N/mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	0.00	138.00	Passed
Pm+Pl (SUS)	29.79	207.00	Passed
Pm+Pl+Q (TOTAL)	347.80	414.00	Passed

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	3836.9998	mm.
Skirt Thickness	STHK	12.0000	mm.
Skirt Internal Corrosion Allowance	SCA	1.0000	mm.
Skirt External Corrosion Allowance		0.0000	mm.
Skirt Material		SA-36	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm.
Design Temperature of the Basing		38.00	°C
Basing Material		SA-516 70	
Basing Ope. All. Stress	BASOPE, f4	138.00	N./mm2
Basing Yield Stress		262.00	N./mm2
Inside Diameter of Basing	DI, D4	3657.0000	mm.
Outside Diameter of Basing	DOU, D5	4117.0000	mm.
Nominal Diameter of Bolts	BND, d(b0)	42.0000	mm.
Bolt Corrosion Allowance	BCA	0.0000	mm.
Root Area of a Single Bolt	Area	8.6100	cm2
Bolt Material		SA-193 B7	
Bolt Material Class/Thickness Range		<= 64	
Bolt Operating Allowable Stress	SA, fB	138.00	N./mm2
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	3976.9998	mm.
Bolt Allowable Shear Stress		82.800	N./mm2
Ultimate Comp. Strength of Concrete	FPC	41.0	N./mm2
Allowable Comp. Strength of Concrete	FC	22.5	N./mm2
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	16.0000	mm.
Width of Gussets at Top Plate	TWDT	130.0000	mm.
Width of Gussets at Base Plate	BWDT	130.0000	mm.
Gusset Plate Elastic Modulus	E	201734000.0	KPa.
Gusset Plate Yield Stress	SY	262.0	N./mm2
Height of Chair	h1	300.0000	mm.
Distance between Gussets	RG, b6	90.0000	mm.
Dist. from Bolt Center to Gusset (Rg/2)	CG	45.0000	mm.
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	38.0000	mm.
Radial Width of the Top Plate	TOPWTH, b5	140.0000	mm.
Anchor Bolt Hole Dia. in Top Plate	dia,H	58.0000	mm.
External Corrosion Allowance	CA	1.0000	mm.
Dead Weight of Vessel	DW	36155.1	Kgf
Operating Weight of Vessel	ROW	50572.3	Kgf
Earthquake Moment on Basing	EQMOM	42638.0	Kg-m.
Wind Moment on Basing	WIMOM	11092.0	Kg-m.
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \text{ per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 42638/3977.0) - 48986) / 20$$



= -305.1 Kgf

Bolt Load, Windload Empty Case:

= $((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3

= $((4 \cdot 11092/3977.0) - 36155) / 20$

= -1250.0 Kgf

Bolt Load, Windload Vortex Shedding Case:

= $((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3

= $((4 \cdot 0/3977.0) - 50572) / 20$

= -2528.6 Kgf

Governing Bolt Load Condition, **** No Tensile Bolt Loads ****

Area Available in one Bolt Abs : 8.6100 cm²

Area Available in all the Bolts Abs * RN : 172.2000 cm²



*There seems to be no tensile load on bolts per the Brownell and Young Method.
It is possible that this method is encountering a problem. It is strongly
recommended that you try the simplified analysis method for the basing.*

Check the Bearing Stress in the Concrete [fc(max)]

= $fc[(2kd + t3) / (2kd)]$

= $0.338[(2 \cdot 1.0 \cdot 3977.0 + 230.0) / (2 \cdot 1.0 \cdot 3977.0)]$

= 0.348 N./mm², Should be ≤ 22.5 N./mm²

Values for table 10.3, l = 140.0, b = 312.353, l/b = 0.448212

Maximum Moment per unit width [Mmax]:

= max(Mx, My) = max(78.284, 245.341) = 245.341 Kgf

Reqd Thickness of Basing, Brownell & Young Method [T]:

= $\sqrt{(6 \cdot Mmax / fallow)} + Ca$

= $\sqrt{(6 \cdot 245.341 / 172.9)} + 1.0$

= 10.137 mm.

Nomenclature:

a = (Dc-Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abs	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 140.0, b = 90.0, b/l = 0.642857

As b/l (0.643) is less than 1, inverting b/l = 1.556.

Moment Term, based on geometry [Mo]:

= $P / (4 \cdot \pi) [1.3(\ln((2 \sin(\pi \cdot a/l) / (\pi \cdot g))) + 1) - [(0.7 - g2)P / (4\pi)]$

= $12115.84 / (4 \cdot \pi) [1.3(\ln((2 \cdot 140.0 \cdot \sin(\pi \cdot 70.0 / 140.0)) /$

$(\pi \cdot 34.925))) + 1) - [(0.7 - 0.065) \cdot 12115.84 / (4 \cdot \pi)]$

= 1525.7030 Kgf

Required Thickness of Continuous Top Ring [Tc]:

= $\sqrt{(6 \cdot Abs(Mo) / Fb)} + Ca$

= $\sqrt{(6 \cdot Abs(1525.7) / 174.66)} + 1.0$

= 23.6712 mm.

Required Gusset Plate Thickness [tg]:

= $P / (Stress Term \cdot lavg) + Ca$

= $12115.84 / (124.11 \cdot 130.0) + 1.0$

= 10.525 (not less than 9.525 + 1.0) mm.

Bolt spacing [m]:



$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3977.0/20 = 624.706 \text{ mm.}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76(P \cdot a / (m \cdot (h + tba) \cdot 1.5 \cdot Sktope))^{2/3} \cdot r^{1/3} + Ca$$

$$= 1.76(12116 \cdot 70.0 / (624.71 \cdot 262.0 \cdot 171))^{2/3} \cdot 1918.51^{1/3} + Ca$$

$$= 9.739 + 1.0 = 10.739 \text{ mm.}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 8722.7 / (8.61 \cdot 20)$$

$$= 5.0 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	10.1371	mm.
Actual Basing Thickness as entered by user	32.0000	mm.
Required Thickness of Chair Cap	23.6712	mm.
Actual Top Ring Thickness as entered by user	38.0000	mm.
Required Gusset thickness, + CA	10.5250	mm.
Actual Gusset Thickness as entered by user	16.0000	mm.
Required Thickness of Skirt for Local Stress	10.7389	mm.
Given Thickness of Skirt	12.0000	mm.
Required Gusset Height to meet local stress	186.2616	mm.

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$$

$$= 7.504 \cdot (12.0 - 1.0)$$

$$= 8.42 \text{ Kg/mm.}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 8 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 0.634 \text{ mm.}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (\text{Cmwth} + 2 \cdot (\text{Hg} + \text{Tta}))$$

$$= 12115.8 / (162.0 + 2 \cdot (230.0 + 38.0))$$

$$= 17.358 \text{ Kg/mm.}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (\text{Cmwth} \cdot (\text{Hg} + \text{Tta}) + 0.6667 \cdot (\text{Hg} + \text{Tta})^2)$$

$$= 12115.8 \cdot 70.0 / (162.0 \cdot (268.0) + 0.6667 \cdot (268.0)^2)$$

$$= 9.289 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{ Wv^2 + Wh^2 }$$

$$= \sqrt{ 17.36^2 + 9.29^2 }$$

$$= 19.687 \text{ Kg/mm.}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 19.69 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 1.484 \text{ mm.}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 12115.8 / (2 \cdot 130.0)$$

$$= 46.599 \text{ Kg/mm.}$$



Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 12115.8 \cdot 70.0 / (2 \cdot 268.0 \cdot 130.0)$$

$$= 12.171 \text{ Kg/mm.}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{46.6^2 + 12.17^2}$$

$$= 48.163 \text{ Kg/mm.}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 48.16 / [(0.4 \cdot 230) \cdot 2 \cdot 0.707]$$

$$= 3.629 \text{ mm.}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basinger to Skirt Double Fillet Weld Size	4.7625	mm.
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm.
Required Top Plate to Skirt Weld Size	6.3500	mm.
Required Gusset to Top Plate Double Fillet Weld Size	3.6295	mm.

Tailing Lug Stress Calculation Results :

Tail Lug Type:	Dual			
Tail lug yield stress	Syt	262.00	N./mm	
Tail lug allowable stress	Sat	138.00	N./mm	
Tail lug thickness	TTL	60.0000	mm.	
Pin hole diameter	DPH	80.0000	mm.	
Fillet weld size	WS	30.0000	mm.	
Tail lug load	TLOAD	9926	Kgf	

Decay Length Along the Skirt (e):

$$= 0.5 \cdot \sqrt{(DS/2 \cdot Ts)}$$

$$= 0.5 \cdot \sqrt{(3837.0/2 \cdot 12.0)}$$

$$= 75.87 \text{ mm.}$$

Skirt Effective Length :

$$= HG + TTA + e$$

$$= 230.0 + 38.0 + 75.87$$

$$= 343.87 \text{ mm.}$$

Section	CG (Y) mm.	Width (H) mm.	Area (A) cm^2	Moment Area mm.^3
Top plate, t	160.00	140.00	53.20	851199.56
Skirt, s	84.00	343.87	41.26	346615.69
Bottom plate, b	115.00	230.00	73.60	846400.75
		TOTAL	168.06	2044215.88

Calculated Overall CG (Y):

$$= M_{total} / A_{total}$$

$$= 2044216 / 168.06$$

$$= 121.63 \text{ mm.}$$

Calculated Moment of Inertia (I):

$$= (At \cdot Yt^2 + Topwth^3 \cdot TTA/12 + As \cdot Ys^2 + Ts^3 \cdot Hs/12 + Ab \cdot Yb^2 + Hb^3 \cdot Tb/12) - M_{total}^2 / A_{total}$$

$$= (53.2 \cdot 160.0^2 + 140.0^3 \cdot 38.0/12 + 41.26 \cdot 84.0^2 + 12.0^3 \cdot 343.87/12 + 73.6 \cdot 115.0^2 + 230.0^3 \cdot 32.0/12) - 2044216^2 / 168.06$$

$$= 5518.31 \text{ cm}^4$$



$$\begin{aligned}\alpha &= \arcsin(OS / (DS / 2)) \\ &= \arcsin(700.0/(3837.0/2)) \\ &= 0.373 \text{ rad. (21.399 deg.)}\end{aligned}$$

$$\begin{aligned}K &= \sin(\alpha) \cdot (\sin(\alpha) - \pi + \alpha) + k2 \cdot (1 + \cos(\alpha)) \\ &= \sin(0.37) \cdot (\sin(0.37) - 3.14 + 0.37) + 1.0 \cdot (1 + \cos(0.37)) \\ &= 1.05\end{aligned}$$

$$\begin{aligned}\text{Required Section Modulus (Zr) - per Roark's Table 17 ref 4} \\ &= K \cdot TLOAD \cdot (DI/2 + Y) / (0.66 \cdot Sy \cdot \pi) \\ &= 1.05 \cdot 9926.6 \cdot (3657.0/2 + 121.63) / (0.66 \cdot 262.0 \cdot 3.142) \\ &= 368406.47 \text{ mm.}^3\end{aligned}$$

$$\begin{aligned}\text{Calculated Section Modulus (Zc):} \\ &= I / Y \\ &= 5518.31/121.63 \\ &= 453684.75 \text{ mm.}^3\end{aligned}$$

$$\begin{aligned}\text{Allowable Shear at the Pin Hole (Ssa):} \\ &= 0.4 \cdot Syt \\ &= 0.4 \cdot 262.0 \\ &= 104.80 \text{ N./mm}^2\end{aligned}$$

$$\begin{aligned}\text{Tail Lug Radius (R):} \\ &= (TBA + HG + TTA) / 2 \\ &= (32.0 + 230.0 + 38.0) / 2 \\ &= 150.00 \text{ mm.}\end{aligned}$$

$$\begin{aligned}\text{Calculated Shear at the Pin Hole (Ssc):} \\ &= TLOAD / (2 \cdot TTL \cdot (R - DPH / 2)) \\ &= 9926.6 / (2 \cdot 60.0 \cdot (150.0 - 80.0/2)) \\ &= 7.37 \text{ N./mm}\end{aligned}$$

$$\begin{aligned}\text{Allowable Bearing Stress at the Pin Hole (Sba):} \\ &= 0.75 \cdot Syt \\ &= 0.75 \cdot 262.0 \\ &= 196.50 \text{ N./mm}^2\end{aligned}$$

$$\begin{aligned}\text{Calculated Bearing Stress at the Pin Hole (Sbc):} \\ &= TLOAD / (DPH \cdot TTL) \\ &= 9926.6 / (80.0 \cdot 60.0) \\ &= 20.28 \text{ N./mm}^2\end{aligned}$$

$$\begin{aligned}\text{Weld Length (LW):} \\ &= 2 \cdot (TOPWTH + TBA + TTA + HG + (DOU - DS) / 2) \\ &= 2 \cdot (140.0 + 32.0 + 38.0 + 230.0 + (4117.0 - 3837.0)/2) \\ &= 1143.00 \text{ mm.}\end{aligned}$$

$$\begin{aligned}\text{Allowable Weld Stress (Swa):} \\ &= 0.49 \cdot Sat \\ &= 0.49 \cdot 138.0 \\ &= 67.62 \text{ N./mm}^2\end{aligned}$$

$$\begin{aligned}\text{Calculated Weld Stress (Swc):} \\ &= TLOAD / (0.707 \cdot WS \cdot LW) \\ &= 9926.6 / (0.707 \cdot 30.0 \cdot 1143.0) \\ &= 4.02 \text{ N./mm}^2\end{aligned}$$

Tailing Lug Results Summary (mm.³ & N./mm²):

	Actual	Allowable or Required	Pass/Fail
Section Modulus	453684.75	368406.47	Passed
Shear Stress	7.37	104.80	Passed
Bearing Stress	20.28	196.50	Passed
Weld Stress	4.02	67.62	Passed

[Review notes about nozzle loadings and supports in the Warnings report.](#)

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Basing Calculations: Step: 25 2:27pm Sep 16,2026

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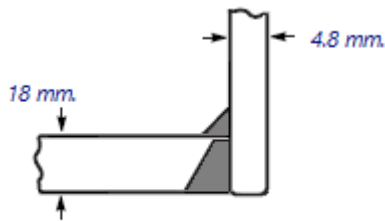

Input, Nozzle Desc: D1-2"
From: 30

Pressure for Reinforcement Calculations	P	6.156	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516	70
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	3806.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	18.0000	mm.
Head Internal Corrosion Allowance	cas	1.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312	TP316L
Material UNS Number			S31603
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	115.00	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80S	
Flange Material		SA-182	F316L
Flange Type		Weld Neck	Flange
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 2.3	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2200.0	mm.
Centerline Distance to Tangent		1300.0	mm.

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: D1-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.993 in.
Actual Thickness Used in Calculation	0.191 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E \cdot 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 6.16 \cdot 3808.0 \cdot 0.8995 / (2 \cdot 138.0 \cdot 1.0 \cdot 0.2 \cdot 6.16)$
 $= 7.6436 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.16 \cdot 25.32 / (115 \cdot 1.0 \cdot 0.6 \cdot 6.16)$
 $= 0.1360 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	101.2698 mm.
Parallel to Vessel Wall, opening length	d	50.6349 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	12.1126 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: D1-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1360 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.4912 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.4912 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 3.4200 mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[3.42, \max(9.4912, 2.5)]$
 $= 3.4200 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$
 $= \max(0.136, 3.42)$
 $= 3.4200 \text{ mm.}$

Available Nozzle Neck Thickness = 4.8450 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	51.5,	Allowable	:	115.0 N./mm2	Passed
Expansion	:	0.0,	Allowable	:	236.0 N./mm2	Passed
Occasional	:	1.5,	Allowable	:	152.9 N./mm2	Passed



Shear : 34.8, Allowable : 80.5 N./mm2 Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: D1-2"

Intermediate Calc. for nozzle/shell Welds Tmin 4.8450 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	3.3915 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1322 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 168.1322 mm.

Input Echo, WRC107/537 Item 1, Description: D1-2" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	1.0000	mm.
Vessel Diameter	Dv	6850.800	mm.
Vessel Thickness	Tv	18.000	mm.
Design Temperature	T1	82.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm2
Vessel Hot S.I. Allowable	Smh	138.00	N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	0.0000	mm.
Nozzle Diameter	Dn	50.635	mm.
Nozzle Thickness	Tn	4.845	mm.
Nozzle Material		SA-312 TP316L	
Nozzle UNS Number		S31603	
Nozzle Cold S.I. Allowable	SNmc	115.00	N./mm2
Nozzle Hot S.I. Allowable	SNmh	115.00	N./mm2
Design Internal Pressure	Dp	6.156	bars
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	382.4	Kgf
Longitudinal Shear (SUS)	(V1) V1	254.9	Kgf
Circumferential Shear (SUS)	(Vc) V2	254.9	Kgf
Circumferential Moment (SUS)	(Mc) M1	35.7	Kg-m.
Longitudinal Moment (SUS)	(Ml) M2	35.7	Kg-m.
Torsional Moment (SUS)	Mt	51.0	Kg-m.

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
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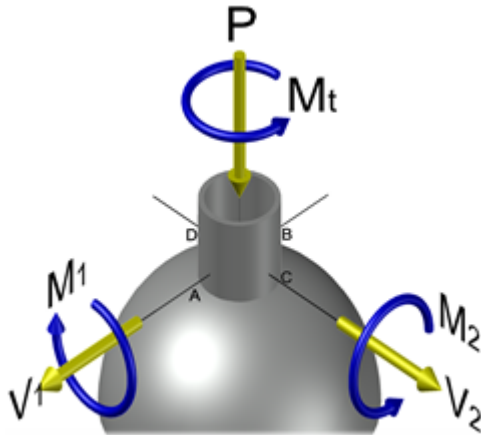
Compute Pressure Stress per WRC-368

No

Local Loads applied at end of Nozzle/Attachment

No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Error/Warning Messages for WRC 107/537 Analysis: 1

Warning - The ratio Ro/Rm, 0.0088, must be > 0.0571 and < 0.571.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 60.325 + 2 \cdot 1.65 \cdot \sqrt{(3434.9 (18.0 - 1.0))}$$

$$= 857.760 \text{ mm.}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load	P	382.4	Kgf
Circumferential Shear	(VC) V2	254.9	Kgf
Longitudinal Shear	(VL) V1	254.9	Kgf
Circumferential Moment	(MC) M1	35.7	Kg-m.
Longitudinal Moment	(ML) M2	35.7	Kg-m.
Torsional Moment	MT	51.0	Kg-m.

Dimensionless Parameters: U = 0.12 TAU = 5.73 RHO = 3.51

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 3	0.11983	(A,B,C,D)
$M(x) / P$	SP 3	0.06599	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.46931	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 3	0.27863	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.46931	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 3	0.27863	(A,B,C,D)
$N(y) * T / P$	SP 3	0.30271	(A,B,C,D)
$M(y) / P$	SP 3	0.24930	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.09806	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 3	1.34429	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.09806	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 3	1.34429	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (N./mm2)



Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-1.6	-1.6	-1.6	-1.6	-1.6	-1.6	-1.6	-1.6
Rad.	Bend. P	-5.1	5.1	-5.1	5.1	-5.1	5.1	-5.1	5.1
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-2.4	-2.4	2.4	2.4
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-8.4	8.4	8.4	-8.4
Rad.	Memb. ML	-2.4	-2.4	2.4	2.4	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-8.4	8.4	8.4	-8.4	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-17.4	9.6	4.0	-2.4	-17.4	9.6	4.0	-2.4
Tang.	Memb. P	-3.9	-3.9	-3.9	-3.9	-3.9	-3.9	-3.9	-3.9
Tang.	Bend. P	-19.4	19.4	-19.4	19.4	-19.4	19.4	-19.4	19.4
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-40.4	40.4	40.4	-40.4
Tang.	Memb. ML	-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-40.4	40.4	40.4	-40.4	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-64.2	55.4	17.6	-24.4	-64.2	55.4	17.6	-24.4
	Shear VC	1.6	1.6	-1.6	-1.6	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-1.6	-1.6	1.6	1.6
	Shear MT	5.1	5.1	5.1	5.1	5.1	5.1	5.1	5.1
Tot.	Shear	6.7	6.7	3.6	3.6	3.6	3.6	6.7	6.7
	Str. Int.	65.2	56.4	18.5	25.0	64.5	55.7	20.3	26.3

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (N./mm²)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	62.0	62.0	62.0	62.0	62.0	62.0	62.0	62.0
Rad.	Pl (SUS)	-3.9	-3.9	0.8	0.8	-3.9	-3.9	0.8	0.8
Rad.	Q (SUS)	-13.5	13.5	3.2	-3.2	-13.5	13.5	3.2	-3.2
Long.	Pm (SUS)	62.0	62.0	62.0	62.0	62.0	62.0	62.0	62.0
Long.	Pl (SUS)	-4.4	-4.4	-3.4	-3.4	-4.4	-4.4	-3.4	-3.4
Long.	Q (SUS)	-59.8	59.8	21.0	-21.0	-59.8	59.8	21.0	-21.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	1.6	1.6	-1.6	-1.6	-1.6	-1.6	1.6	1.6
Shear	Q (SUS)	5.1	5.1	5.1	5.1	5.1	5.1	5.1	5.1
Pm (SUS)		62.0	62.0	62.0	62.0	62.0	62.0	62.0	62.0
Pm+Pl (SUS)		59.4	59.4	63.3	63.3	59.4	59.4	63.3	63.3
Pm+Pl+Q (Total)		48.7	118.4	80.5	60.2	47.4	117.7	82.4	61.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	62.04	138.00	Passed
Pm+Pl (SUS)	63.34	207.00	Passed
Pm+Pl+Q (TOTAL)	118.40	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Nozzle Calcs.: D1-2"



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Nozl: 29 2:27pm Sep 16,2026



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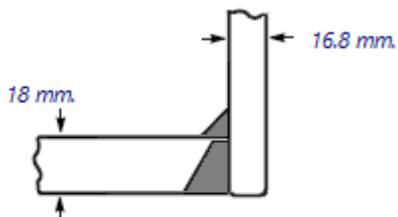
**Input, Nozzle Desc: K1-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K1-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 1904.0 / (138 \cdot 1.0 \cdot 0.6 \cdot 6.0)$
 $= 8.3004 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.0 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1509 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	118.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	59.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1509 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.3004 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.3004 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(9.3004, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1509, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K1-2"

Intermediate Calc. for nozzle/shell welds	Tmin	16.7640 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4672 mm.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Nozzle Calcs.: K1-2" Nozl: 30 2:27pm Sep 16,2026

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HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.8423 mm.

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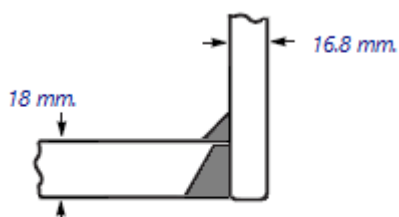
**Input, Nozzle Desc: K3-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		80.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K3-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 1904.0 / (138 \cdot 1.0 \cdot 0.6 \cdot 6.0)$
 $= 8.3004 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.0 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1509 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	118.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	59.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1509 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.3004 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.3004 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(9.3004, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1509, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K3-2"

Intermediate Calc. for nozzle/shell welds	Tmin	16.7640 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4672 mm.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Nozzle Calcs.: K3-2"



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Nozl: 31 2:27pm Sep 16,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.8423 mm.

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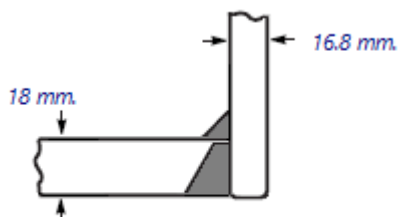

Input, Nozzle Desc: K5-2"
From: 40

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		100.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: K5-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----

Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 1904.0 / (138 \cdot 1.0 \cdot 0.6 \cdot 6.0)$
 $= 8.3004 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.0 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1509 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	118.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	59.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1509 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.3004 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.3004 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(9.3004, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1509, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K5-2"

Intermediate Calc. for nozzle/shell welds	Tmin	16.7640 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4672 mm.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Nozzle Calcs.: K5-2" Nozl: 32 2:27pm Sep 16,2026

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HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.8423 mm.

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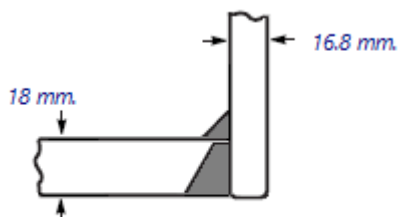
**Input, Nozzle Desc: K7-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K7-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E \cdot 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 1904.0 / (138 \cdot 1.0 \cdot 0.6 \cdot 6.0)$
 $= 8.3004 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.0 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1509 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	118.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	59.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K7-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1509 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.3004 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.3004 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(9.3004, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1509, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K7-2"

Intermediate Calc. for nozzle/shell Welds	Tmin	16.7640 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4672 mm.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

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Nozzle Calcs.: K7-2"

Nozl: 33 2:27pm Sep 16,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.8423 mm.

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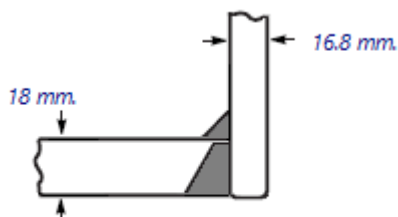
**Input, Nozzle Desc: K2-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		80.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K2-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation

0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.05 \cdot 1904.0 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$
 $= 8.3678 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.05 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1521 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	118.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	59.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1521 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.3678 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.3678 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(9.3678, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1521, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K2-2"

Intermediate Calc. for nozzle/shell welds	Tmin	16.7640 mm.
---	------	-------------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4672 mm.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Nozzle Calcs.: K2-2" Nozl: 34 2:27pm Sep 16,2026

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The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.8423 mm.

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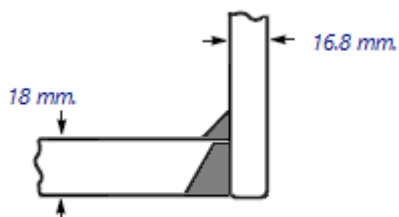
**Input, Nozzle Desc: K4-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm2
Allowable Stress At Ambient	Sna	115.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K4-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation

0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $6.05 \cdot 1904.0 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$

= 8.3678 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)

= $25.4 \cdot (\exp([6.05 / (101.29 \cdot 1.0)] - 1))$

= 0.1521 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 118.3280 mm.

Parallel to Vessel Wall Rn+tn+t 59.1640 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.9100 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K4-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 0.1521 mm.

Wall Thickness per UG16(b), tr16b = 2.5000 mm.

Wall Thickness, shell/head, internal pressure trb1 = 9.3678 mm.

Wall Thickness tb1 = max(trb1, tr16b) = 9.3678 mm.

Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm.

Wall Thickness per table UG-45 tb3 = 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[4.8, max(9.3678, 2.5)]

= 4.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(0.1521, 4.8)

= 4.8000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K4-2"

Intermediate Calc. for nozzle/shell Welds Tmin 16.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4672 mm.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Nozzle Calcs.: K4-2"

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Nozl: 35 2:27pm Sep 16,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.8423 mm.

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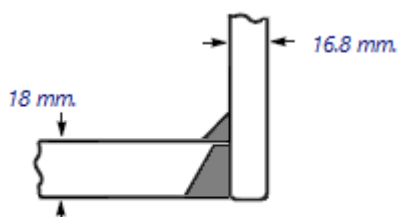
**Input, Nozzle Desc: K6-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		100.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K6-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation

0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $6.05 \cdot 1904.0 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$

= 8.3678 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)

= $25.4 \cdot (\exp([6.05 / (101.29 \cdot 1.0)] - 1))$

= 0.1521 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 118.3280 mm.

Parallel to Vessel Wall Rn+tn+t 59.1640 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.9100 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K6-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 0.1521 mm.

Wall Thickness per UG16(b), tr16b = 2.5000 mm.

Wall Thickness, shell/head, internal pressure trb1 = 9.3678 mm.

Wall Thickness tb1 = max(trb1, tr16b) = 9.3678 mm.

Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm.

Wall Thickness per table UG-45 tb3 = 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[4.8, max(9.3678, 2.5)]

= 4.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(0.1521, 4.8)

= 4.8000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K6-2"

Intermediate Calc. for nozzle/shell Welds Tmin 16.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4672 mm.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Nozzle Calcs.: K6-2"



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Nozl: 36 2:27pm Sep 16,2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.8423 mm.

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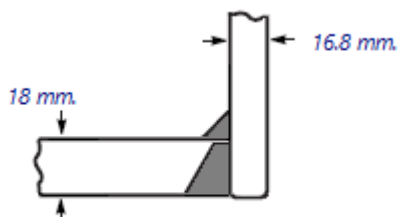
**Input, Nozzle Desc: K8-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K8-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation

0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $6.05 \cdot 1904.0 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$

= 8.3678 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)

= $25.4 \cdot (\exp([6.05 / (101.29 \cdot 1.0)] - 1))$

= 0.1521 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 118.3280 mm.

Parallel to Vessel Wall Rn+tn+t 59.1640 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 41.9100 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K8-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 0.1521 mm.

Wall Thickness per UG16(b), tr16b = 2.5000 mm.

Wall Thickness, shell/head, internal pressure trb1 = 9.3678 mm.

Wall Thickness tb1 = max(trb1, tr16b) = 9.3678 mm.

Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm.

Wall Thickness per table UG-45 tb3 = 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[4.8, max(9.3678, 2.5)]

= 4.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(0.1521, 4.8)

= 4.8000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K8-2"

Intermediate Calc. for nozzle/shell Welds Tmin 16.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4672 mm.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Nozzle Calcs.: K8-2" Nozl: 37 2:27pm Sep 16,2026

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HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.8423 mm.

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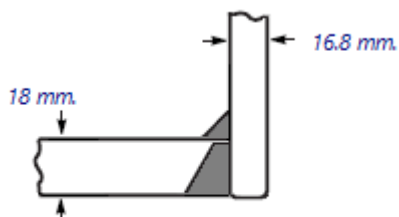
**Input, Nozzle Desc: K10-2"****From: 40**

Pressure for Reinforcement Calculations	P	6.049	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		150.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F316L	
Material UNS Number		S31603	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.29	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-182 F316L	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 2.3	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: K10-2"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
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Actual Thickness Used in Calculation 0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.05 \cdot 1904.0 / (138 \cdot 1.0 - 0.6 \cdot 6.05)$
 $= 8.3678 \text{ mm.}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([6.05 / (101.29 \cdot 1.0)] - 1))$
 $= 0.1521 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	118.3280 mm.
Parallel to Vessel Wall	Rn+tn+t	59.1640 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: K10-2".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.1521 mm.
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 9.3678 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 9.3678 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm.
Wall Thickness per table UG-45	tb3	= 4.8000 mm.

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(9.3678, 2.5)]$
 $= 4.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.1521, 4.8)$
 $= 4.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: K10-2"

Intermediate Calc. for nozzle/shell welds	Tmin	16.7640 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4672 mm.

Process Air 2nd KOD 197-D

PV Elite 28 U1 Licensee: SL User

FileName : Process Air Compressor 2nd KO Drum-197-D-7%

Nozzle Calcs.: K10-2"

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Nozl: 38 2:27pm Sep 16, 2026



HEXAGON

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.8423 mm.

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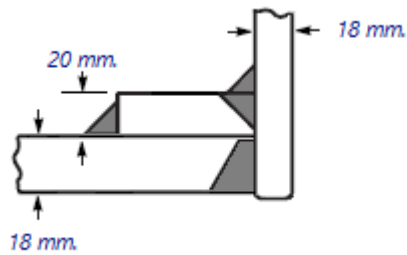
**Input, Nozzle Desc: N1-30"****From: 50**

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1850.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1262.0000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		250.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N1-30"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.709 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 1904.0 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 8.3004 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.0 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 6.0)$
 $= 1.6537 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1464.0000 mm.
Parallel to Vessel Wall, opening length	d	732.0000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.5000 mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr4]:

$$= \min(1, S_p / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1.0, 1.0)$$

$$= 1.000$$

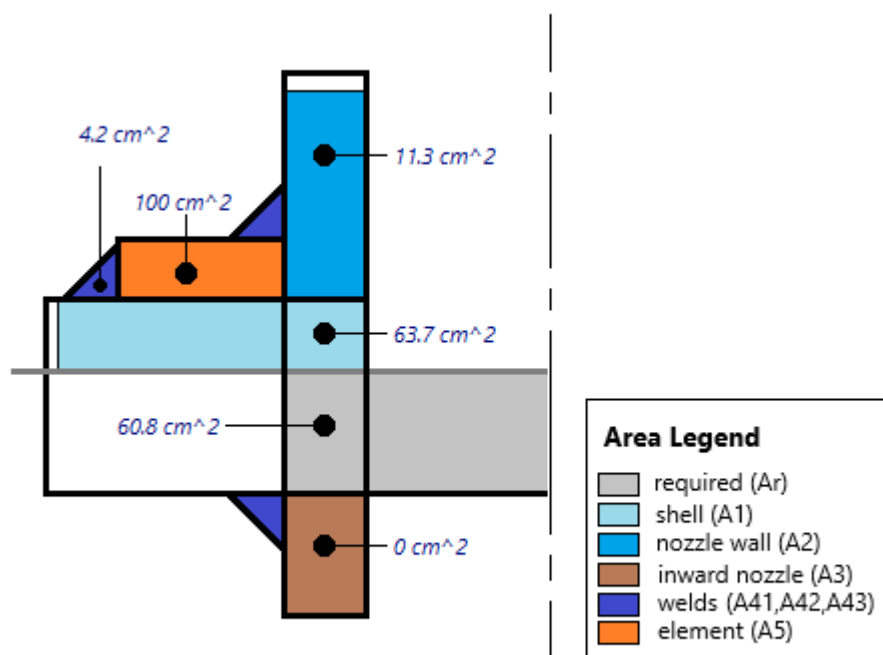


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	60.759	NA	NA
Area in Shell	A1	63.681	NA	NA
Area in Nozzle Wall	A2	11.344	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.240	NA	NA
Area in Element	A5	100.000	NA	NA
TOTAL AREA AVAILABLE	Atot	179.265	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (732.0 \cdot 8.3004 \cdot 1.0 + 2 \cdot 15.0 \cdot 8.3004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 60.759 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 732.0(1.0 \cdot 17.0 - 1.0 \cdot 8.3) - 2 \cdot 15.0$
 $(1.0 \cdot 17.0 - 1.0 \cdot 8.3004) \cdot (1 - 1.0)$
 $= 63.681 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 42.5) \cdot (15.0 - 1.65) \cdot 1.0$
 $= 11.344 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$$

$$= 4.240 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lp}, t_e)) \cdot f_{r4}$$

$$= (1262.0 - 762.0) \cdot 20.0 \cdot 1.0$$

$$= 100.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.6537 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 9.3004 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 9.3004 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.331, \max(9.3004, 4.5)]$$

$$= 9.3004 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.6537, 9.3004)$$

$$= 9.3004 \text{ mm.}$$

Available Nozzle Neck Thickness = 18.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.0,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	319.0 N./mm ²	Passed
Occasional	:	7.2,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	12.0,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 18.0$, $tr = 1.654$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 18.0$, $tr = 1.654$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 18.0$, $tr = 1.654$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 18.0, tr = 8.3, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.488$, Temp. Reduction = 34 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 8.3, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.488$, Temp. Reduction = 34 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -44 °C
 Gov. MDMT of the Reinforcement Pad : -44 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.00/19.60 = 0.306$

Weld Size Calculations, Description: N1-30"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 17.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$8.5000 = 0.5 \cdot \text{TminPad}$	$12.7260 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (60.7589 - 63.6811 + 2 \cdot 15.0 \cdot 1.0 \cdot (1.0 \cdot 17.0 - 8.3004)) \cdot 138)$
 $= \max(0, -439.47) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (11.3443 + 100.0 + 4.24 - 0.0 \cdot 1.0) \cdot 138$
 $= 162648.23 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (11.3443 + 0.0 + 1.0 + (5.1)) \cdot 138$
 $= 24547.34 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (11.3443 + 0.0 + 4.24 + 100.0 + (5.1)) \cdot 138$
 $= 169824.86 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 762.0 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 82532. \text{ Kgf}$



Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1262.0 \cdot 18.0 \cdot 0.49 \cdot 138$$

$$= 246036. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 373.5) \cdot (18.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 173372. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 762.0 \cdot 20.0 \cdot 0.74 \cdot 138$$

$$= 249280. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 762.0 \cdot (18.0 - 1.0) \cdot 0.74 \cdot 138$$

$$= 211888. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (246036 + 173373) = 419408 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (82532 + 249280 + 211888 + 0) = 543699 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (246036 + 211888 + 0) = 457924 \text{ Kg}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 419408 \text{ Kg}, \text{ must exceed } W = 0 \text{ Kg} \quad \text{or } W1 = 162648 \text{ Kg}$$

$$\text{Path 2-2} = 543699 \text{ Kg}, \text{ must exceed } W = 0 \text{ Kg} \quad \text{or } W2 = 24547 \text{ Kg}$$

$$\text{Path 3-3} = 457923 \text{ Kg}, \text{ must exceed } W = 0 \text{ Kg} \quad \text{or } W3 = 169824 \text{ Kg}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 38.5302 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 356.5303 mm.

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) \quad 2.419 \%$$

Input Echo, WRC297 Item 1, Description: N1-30" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	3806.000	mm.
Shell Thickness	18.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.



Nozzle Diameter	762.000	mm.
Nozzle Thickness	18.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1262.000	mm.
Thickness of Reinforcing Pad	20.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	6.00	bars
Radial Load	P	5736.27	Kgf
Circumferential Shear	Vc	3824.19	Kgf
Longitudinal Shear	Vl	3824.19	Kgf
Circumferential Moment	Mc	8027.71	Kg-m.
Longitudinal Moment	Ml	8027.71	Kg-m.
Torsional Moment	Mt	11468.10	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 762.0 + 2 \cdot 1.65 \cdot \sqrt{(1922.5(18.0 - 1.0))} \\
 &= 1358.584 \text{ mm.}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3845.000	mm.
Nozzle Outside Diameter	(d) =	762.000	mm.
Vessel Thickness used	(T) =	37.000	mm.
Nozzle Thickness used	(t) =	15.000	mm.
T / t	=	2.467	
d / t	=	50.800	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.020$$

Nr/P	=	0.043
Mr/P	=	0.078
M0/P	=	0.049
N0/P	=	0.165
MrD/Mc	=	0.169
NrDL/Mc	=	0.086
M0d/Mc	=	0.129
N0DL/Mc	=	0.246
MrD/Ml	=	0.092
NrDL/Ml	=	0.053
M0D/Ml	=	0.060
N0DL/Ml	=	0.334

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-12	12	-12	12
Outplane Membrane (Mc)	0	0	0	0



Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-25	-25	25	25
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	30	31	30	31

Outplane Stress Summary	-40	38	64	34
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-19	19	-19	19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-41	41	41	-41
Inplane Pressure Stress	15	15	15	15

Inplane Stress Summary	-51	70	40	-5
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	52	70	64	39
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-12	12	-12	12
Outplane Membrane (Mc)	-18	-18	18	18
Outplane Bending (Mc)	-58	58	58	-58
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	15	15	15	15

Outplane Stress Summary	-80	60	73	-19
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-19	19	-19	19
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-76	76	76	-76
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	30	31	30	31

Inplane Stress Summary	-73	118	92	-21
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**Vessel Stresses (N./mm²):**

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		3	3	3	3
Shear Stress Summary		2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		81	118	93	24

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the hoop direction)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)		-6	-6	-6	-6
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		-25	-25	25	25
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		17	17	17	17
Hoop Stress Summary		-14	-14	35	36

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)		-1	-1	-1	-1
Axial Bending (P)		-84	84	-84	84
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		-12	-12	12	12
Axial Bending (ML)		-181	181	181	-181
Axial Pressure Stress		8	8	8	8
Axial Stress Summary		-270	260	116	-77

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		2	2	-2	-2
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		8	8	8	8
Shear Stress Summary		10	10	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Outside	Inside	Outside	Inside
Two * Max Shear Stress		271	275	116	114

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses in the hoop direction)		Left Outside	Left Inside	Right Outside	Right Inside



Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	17	17	17
Hoop Stress Summary	-8	-7	28	29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-84	84	-84	84
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-345	345	345	-345
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-435	424	280	-242

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8
Shear Stress Summary	6	6	10	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	435	432	281	272

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 762.0 + 2 \cdot 1.65 \cdot \sqrt{1912.5(18.0 - 1.0)}$$

$$= 1357.030 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3825.000 mm.
Nozzle Outside Diameter	(d) =	1262.000 mm.
Vessel Thickness used	(T) =	17.000 mm.
Nozzle Thickness used	(t) =	265.000 mm.
T / t	=	0.064
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.949$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.046
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.025
MrD/Mc	=	0.240
NrDL/Mc	=	0.101



$M\theta d/Mc = 0.072$
 $N\theta DL/Mc = 0.049$
 $MrD/Ml = 0.049$
 $NrDL/Ml = 0.050$
 $M\theta D/Ml = 0.015$
 $N\theta DL/Ml = 0.024$

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	66	67	66	67
Outplane Stress Summary	15	99	64	70

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-82	82	-82	82
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-63	63	63	-63
Inplane Pressure Stress	33	33	33	33
Inplane Stress Summary	-132	159	15	54

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	147	160	64	71

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-92	92	92	-92
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	33	33	33	33
Outplane Stress Summary	-96	133	109	-31

**Vessel Stresses (N./mm²):**

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-82	82	-82	82
Inplane Membrane (Mc)	-21	-21	21	21
Inplane Bending (Mc)	-310	310	310	-310
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	66	67	66	67
Inplane Stress Summary	-357	430	307	-147

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	357	430	307	147

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	31	30	31	30	31	30	31
Circ. Pl (SUS)	-31	-31	19	19	-7	-7	5	5
Circ. Q (SUS)	-39	39	15	-15	-95	95	57	-57
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-4	-4	2	2	-24	-24	12	12
Long. Q (SUS)	-60	60	22	-22	-70	70	46	-46
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	30.0	31.0	30.0	31.0	30.0	31.0	30.0	31.0
Pm+Pl (SUS)	12.0	11.0	49.0	50.0	32.0	33.0	35.0	36.0
Pm+Pl+Q (Total)	49.9	71.3	64.4	40.4	80.1	119.2	92.5	23.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	31.00	138.00	Passed
Pm+Pl (SUS)	50.00	207.00	Passed
Pm+Pl+Q (TOTAL)	119.15	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	66	67	66	67	66	67	66	67
Circ. Pl (SUS)	-9	-9	1	1	-29	-29	13	13
Circ. Q (SUS)	-41	41	-3	3	-392	392	228	-228
Long. Pm (SUS)	33	33	33	33	33	33	33	33
Long. Pl (SUS)	-18	-18	2	2	-14	-14	6	6
Long. Q (SUS)	-145	145	-19	19	-114	114	70	-70
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	66.0	67.0	66.0	67.0	66.0	67.0	66.0	67.0
Pm+Pl (SUS)	57.0	58.0	67.0	68.0	37.1	38.1	79.0	80.0
Pm+Pl+Q (Total)	146.1	160.1	64.0	71.1	355.0	430.0	307.0	148.1

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	67.00	138.00	Passed
Pm+Pl (SUS)	80.02	207.00	Passed
Pm+Pl+Q (TOTAL)	430.00	414.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	17	17	17	17	17	17	17
Circ. Pl (SUS)	-31	-31	19	19	-24	-24	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-13	-13	11	11	-13	-13	11	11
Long. Q (SUS)	-265	265	97	-97	-429	429	261	-261
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	17.0	17.0	17.0	17.0	17.0	17.0	17.0	17.0
Pm+Pl (SUS)	14.4	14.4	36.2	36.2	8.2	8.2	29.4	29.4
Pm+Pl+Q (Total)	270.4	274.7	116.4	114.6	434.1	431.2	280.4	271.7

Nozzle Stress Summation Comparison (N./mm2):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	138.00	Passed
Pm+Pl (SUS)	36.23	207.00	Passed
Pm+Pl+Q (TOTAL)	434.08	414.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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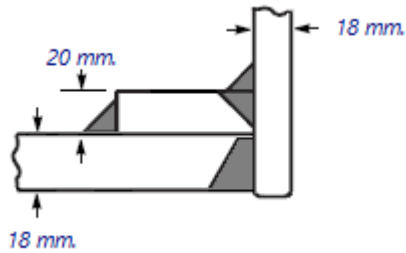
**Input, Nozzle Desc: N2-30"****From: 50**

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2450.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1262.0000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		250.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: N2-30"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.709 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 1904.0 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 8.3004 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.0 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 6.0)$
 $= 1.6537 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1464.0000 mm.
Parallel to Vessel Wall, opening length	d	732.0000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.5000 mm.

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

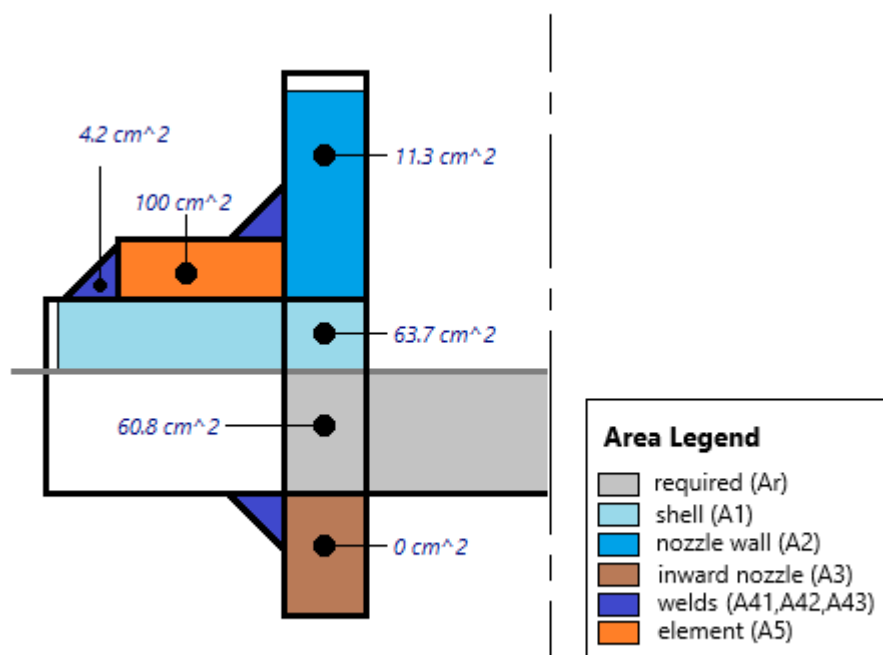


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	60.759	NA	NA
Area in Shell	A1	63.681	NA	NA
Area in Nozzle Wall	A2	11.344	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.240	NA	NA
Area in Element	A5	100.000	NA	NA
TOTAL AREA AVAILABLE	Atot	179.265	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (732.0 \cdot 8.3004 \cdot 1.0 + 2 \cdot 15.0 \cdot 8.3004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 60.759 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 732.0(1.0 \cdot 17.0 - 1.0 \cdot 8.3) - 2 \cdot 15.0$
 $(1.0 \cdot 17.0 - 1.0 \cdot 8.3004) \cdot (1 - 1.0)$
 $= 63.681 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 42.5) \cdot (15.0 - 1.65) \cdot 1.0$
 $= 11.344 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$$

$$= 4.240 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_r$$

$$= (1262.0 - 762.0) \cdot 20.0 \cdot 1.0$$

$$= 100.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.6537 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 9.3004 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 9.3004 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.331, \max(9.3004, 4.5)]$$

$$= 9.3004 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.6537, 9.3004)$$

$$= 9.3004 \text{ mm.}$$

Available Nozzle Neck Thickness = 18.0000 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	26.0,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	319.0 N./mm ²	Passed
Occasional	:	7.2,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	12.0,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 18.0$, $tr = 1.654$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 18.0$, $tr = 1.654$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 18.0$, $tr = 1.654$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 18.0, tr = 8.3, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.488$, Temp. Reduction = 34 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 18.0, tr = 8.3, c = 1.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.488$, Temp. Reduction = 34 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -44 °C
 Gov. MDMT of the Reinforcement Pad : -44 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.00/19.60 = 0.306$

Weld Size Calculations, Description: N2-30"

Intermediate Calc. for nozzle/shell Welds Tmin 15.0000 mm.
 Intermediate Calc. for pad/shell Welds TminPad 17.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$8.5000 = 0.5 \cdot \text{TminPad}$	$12.7260 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (60.7589 - 63.6811 + 2 \cdot 15.0 \cdot 1.0 \cdot (1.0 \cdot 17.0 - 8.3004)) \cdot 138)$
 $= \max(0, -439.47) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (11.3443 + 100.0 + 4.24 - 0.0 \cdot 1.0) \cdot 138$
 $= 162648.23 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (11.3443 + 0.0 + 1.0 + (5.1)) \cdot 138$
 $= 24547.34 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (11.3443 + 0.0 + 4.24 + 100.0 + (5.1)) \cdot 138$
 $= 169824.86 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 762.0 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 82532. \text{ Kgf}$



Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1262.0 \cdot 18.0 \cdot 0.49 \cdot 138$$

$$= 246036. \text{ Kg}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 373.5) \cdot (18.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 173372. \text{ Kg}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 762.0 \cdot 20.0 \cdot 0.74 \cdot 138$$

$$= 249280. \text{ Kg}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 762.0 \cdot (18.0 - 1.0) \cdot 0.74 \cdot 138$$

$$= 211888. \text{ Kg}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (246036 + 173373) = 419408 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (82532 + 249280 + 211888 + 0) = 543699 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (246036 + 211888 + 0) = 457924 \text{ Kg}$$

Summary of Failure Path Calculations:

$$\text{Path 1-1} = 419408 \text{ Kg}, \text{ must exceed } W = 0 \text{ Kg} \quad \text{or } W1 = 162648 \text{ Kg}$$

$$\text{Path 2-2} = 543699 \text{ Kg}, \text{ must exceed } W = 0 \text{ Kg} \quad \text{or } W2 = 24547 \text{ Kg}$$

$$\text{Path 3-3} = 457923 \text{ Kg}, \text{ must exceed } W = 0 \text{ Kg} \quad \text{or } W3 = 169824 \text{ Kg}$$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 38.5302 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 356.5303 mm.

Percent Elongation Calculations:

$$\% \text{ Elongation per Table UG-79-1 } (50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)) = 2.419 \%$$

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 385.36 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 732.00 mm.

Input Echo, WRC297 Item 1, Description: N2-30" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	3806.000	mm.
Shell Thickness	18.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	762.000	mm.
Nozzle Thickness	18.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 70	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1262.000	mm.
Thickness of Reinforcing Pad	20.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	6.00	bars
Radial Load	P	5736.27	Kgf
Circumferential Shear	Vc	3824.19	Kgf
Longitudinal Shear	Vl	3824.19	Kgf
Circumferential Moment	Mc	8027.71	Kg-m.
Longitudinal Moment	Ml	8027.71	Kg-m.
Torsional Moment	Mt	11468.10	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 762.0 + 2 \cdot 1.65 \cdot \sqrt{1922.5(18.0 - 1.0)}$$

$$= 1358.584 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3845.000	mm.
Nozzle Outside Diameter	(d) =	762.000	mm.
Vessel Thickness used	(T) =	37.000	mm.
Nozzle Thickness used	(t) =	15.000	mm.
T / t	=	2.467	
d / t	=	50.800	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.020$$

Nr/P	=	0.043
Mr/P	=	0.078
M0/P	=	0.049
N0/P	=	0.165
MrD/Mc	=	0.169
NrDL/Mc	=	0.086
M0d/Mc	=	0.129
N0DL/Mc	=	0.246
MrD/Ml	=	0.092
NrDL/Ml	=	0.053
M0D/Ml	=	0.060
N0DL/Ml	=	0.334

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-12	12	-12	12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-25	-25	25	25
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	30	31	30	31

Outplane Stress Summary	-40	38	64	34
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-19	19	-19	19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-41	41	41	-41
Inplane Pressure Stress	15	15	15	15

Inplane Stress Summary	-51	70	40	-5
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	4	4	2	2
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	52	70	64	39
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	-6	-6	-6	-6
Outplane Bending (P)	-12	12	-12	12
Outplane Membrane (Mc)	-18	-18	18	18
Outplane Bending (Mc)	-58	58	58	-58
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	15	15	15	15

Outplane Stress Summary	-80	60	73	-19
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-1	-1	-1	-1
Inplane Bending (P)	-19	19	-19	19
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-76	76	76	-76
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	30	31	30	31
Inplane Stress Summary	-73	118	92	-21

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	81	118	93	24

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-25	-25	25	25
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	17	17	17
Hoop Stress Summary	-14	-14	35	36

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-84	84	-84	84
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-12	-12	12	12
Axial Bending (ML)	-181	181	181	-181
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-270	260	116	-77

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	10	10	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	271	275	116	114

**Nozzle Stresses (N./mm²):**

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	-6	-6	-6	-6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	17	17	17	17
Hoop Stress Summary	-8	-7	28	29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	-1	-1	-1	-1
Axial Bending (P)	-84	84	-84	84
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-345	345	345	-345
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-435	424	280	-242

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8
Shear Stress Summary	6	6	10	10

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	435	432	281	272

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 762.0 + 2 \cdot 1.65 \cdot \sqrt{(1912.5(18.0 - 1.0))}$$

$$= 1357.030 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 3825.000 mm.

Nozzle Outside Diameter (d) = 1262.000 mm.

Vessel Thickness used (T) = 17.000 mm.

Nozzle Thickness used (t) = 265.000 mm.

T / t = 0.064

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.949$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.046

Mr/P = 0.071



MØ/P = 0.019
 NØ/P = 0.025
 MrD/Mc = 0.240
 NrDL/Mc = 0.101
 MØd/Mc = 0.072
 NØDL/Mc = 0.049
 MrD/Ml = 0.049
 NrDL/Ml = 0.050
 MØD/Ml = 0.015
 NØDL/Ml = 0.024

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	66	67	66	67
Outplane Stress Summary	15	99	64	70

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-82	82	-82	82
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-10	-10	10	10
Inplane Bending (ML)	-63	63	63	-63
Inplane Pressure Stress	33	33	33	33
Inplane Stress Summary	-132	159	15	54

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	147	160	64	71

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	-4	-4	-4	-4
Outplane Bending (P)	-22	22	-22	22
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-92	92	92	-92
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	33	33	33	33

Outplane Stress Summary	-96	133	109	-31
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	-8	-8	-8	-8
Inplane Bending (P)	-82	82	-82	82
Inplane Membrane (Mc)	-21	-21	21	21
Inplane Bending (Mc)	-310	310	310	-310
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	66	67	66	67

Inplane Stress Summary	-357	430	307	-147
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	357	430	307	147
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:**Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):**

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	31	30	31	30	31	30	31
Circ. Pl (SUS)	-31	-31	19	19	-7	-7	5	5
Circ. Q (SUS)	-39	39	15	-15	-95	95	57	-57
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	-4	-4	2	2	-24	-24	12	12
Long. Q (SUS)	-60	60	22	-22	-70	70	46	-46
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	30.0	31.0	30.0	31.0	30.0	31.0	30.0	31.0
Pm+Pl (SUS)	12.0	11.0	49.0	50.0	32.0	33.0	35.0	36.0
Pm+Pl+Q (Total)	49.9	71.3	64.4	40.4	80.1	119.2	92.5	23.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	31.00	138.00	Passed
Pm+Pl (SUS)	50.00	207.00	Passed
Pm+Pl+Q (TOTAL)	119.15	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	66	67	66	67	66	67	66	67
Circ. Pl (SUS)	-9	-9	1	1	-29	-29	13	13
Circ. Q (SUS)	-41	41	-3	3	-392	392	228	-228
Long. Pm (SUS)	33	33	33	33	33	33	33	33
Long. Pl (SUS)	-18	-18	2	2	-14	-14	6	6
Long. Q (SUS)	-145	145	-19	19	-114	114	70	-70
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	66.0	67.0	66.0	67.0	66.0	67.0	66.0	67.0
Pm+Pl (SUS)	57.0	58.0	67.0	68.0	37.1	38.1	79.0	80.0
Pm+Pl+Q (Total)	146.1	160.1	64.0	71.1	355.0	430.0	307.0	148.1

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	67.00	138.00	Passed
Pm+Pl (SUS)	80.02	207.00	Passed
Pm+Pl+Q (TOTAL)	430.00	414.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	17	17	17	17	17	17	17	17
Circ. Pl (SUS)	-31	-31	19	19	-24	-24	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-13	-13	11	11	-13	-13	11	11
Long. Q (SUS)	-265	265	97	-97	-429	429	261	-261
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	17.0	17.0	17.0	17.0	17.0	17.0	17.0	17.0
Pm+Pl (SUS)	14.4	14.4	36.2	36.2	8.2	8.2	29.4	29.4
Pm+Pl+Q (Total)	270.4	274.7	116.4	114.6	434.1	431.2	280.4	271.7

**Nozzle Stress Summation Comparison (N./mm²):**

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	138.00	Passed
Pm+Pl (SUS)	36.23	207.00	Passed
<i>Pm+Pl+Q (TOTAL)</i>	<i>434.08</i>	<i>414.00</i>	<i>Failed</i>

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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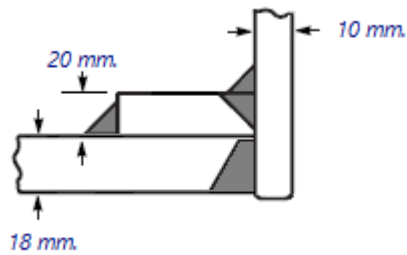
**Input, Nozzle Desc: M1-24"****From: 50**

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1600.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	1009.6000	mm.
Thickness of Pad	te	20.0000	mm.
Weld leg size between Pad and Shell	Wp	18.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm.
Reinforcing Pad Width		200.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).

**Reinforcement CALCULATION, Description: M1-24"**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.0 \cdot 1904.0 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$
 $= 8.3004 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.0 \cdot 304.8 / (138 \cdot 1.0 + 0.4 \cdot 6.0)$
 $= 1.3230 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1191.2001 mm.
Parallel to Vessel Wall, opening length	d	595.6000 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr4]:

$$= \min(1, S_p / S_v)$$

$$= \min(1, 138.0 / 138.0)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1.0, 1.0)$$

$$= 1.000$$

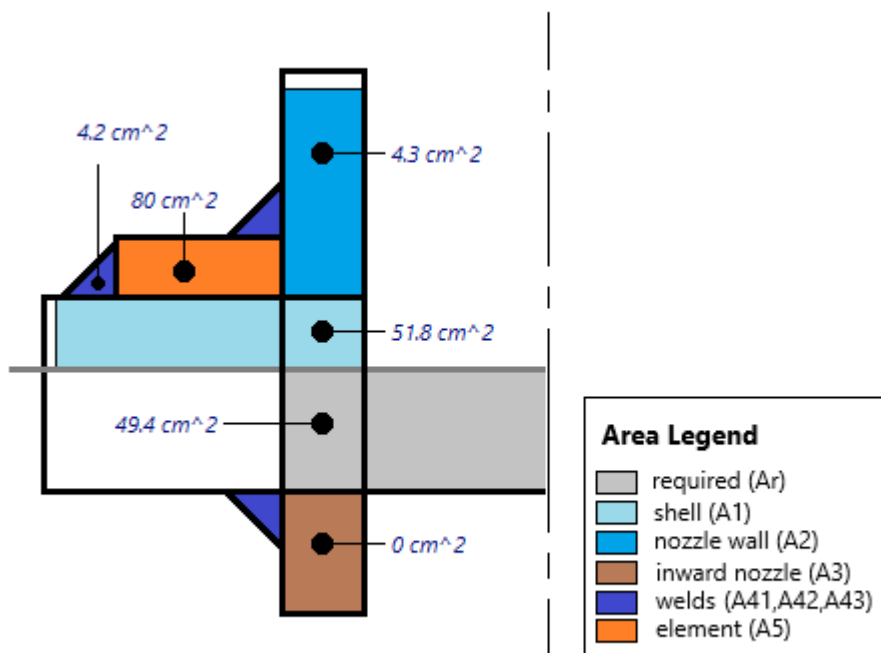


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	49.437	NA	NA
Area in Shell	A1	51.815	NA	NA
Area in Nozzle Wall	A2	4.258	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	4.240	NA	NA
Area in Element	A5	80.000	NA	NA
TOTAL AREA AVAILABLE	Atot	140.313	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (595.6 \cdot 8.3004 \cdot 1.0 + 2 \cdot 7.0 \cdot 8.3004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 49.437 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 595.6(1.0 \cdot 17.0 - 1.0 \cdot 8.3) - 2 \cdot 7.0$
 $(1.0 \cdot 17.0 - 1.0 \cdot 8.3004) \cdot (1 - 1.0)$
 $= 51.815 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (7.0 - 1.32) \cdot 1.0$
 $= 4.258 \text{ cm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi - can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$



$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 18.0^2 \cdot 1.0$$

$$= 4.240 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot f_{r4}$$

$$= (1009.6001 - 609.6) \cdot 20.0 \cdot 1.0$$

$$= 80.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:Wall Thickness for Internal/External pressures $t_a = 4.3230 \text{ mm}$.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$

$$= 4.3230 \text{ mm}$$

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)**

Govrn. thk, $t_g = 10.0$, $t_r = 1.323$, $c = 3.0 \text{ mm}$., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.189$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10.0$, $t_r = 1.323$, $c = 3.0 \text{ mm}$., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.189$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 10.0$, $t_r = 1.323$, $c = 3.0 \text{ mm}$., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.189$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 18.0$, $t_r = 8.3$, $c = 1.0 \text{ mm}$., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.488$, Temp. Reduction = 34 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 10.0$, $t_r = 1.323$, $c = 3.0 \text{ mm}$., $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.189$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -44 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -44 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.00/51.00 = 0.118$

Weld Size Calculations, Description: M1-24"

Intermediate Calc. for nozzle/shell Welds $T_{min} = 7.0000$ mm.
 Intermediate Calc. for pad/shell Welds $T_{minPad} = 17.0000$ mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$8.5000 = 0.5 * T_{minPad}$	$12.7260 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (49.4372 - 51.8148 + 2 \cdot 7.0 \cdot 1.0 \cdot (1.0 \cdot 17.0 - 8.3004)) \cdot 138)$
 $= \max(0, -1631.95) \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (4.2578 + 80.0 + 4.24 - 0.0 \cdot 1.0) \cdot 138$
 $= 124532.50 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (4.2578 + 0.0 + 1. + (2.38)) \cdot 138$
 $= 10747.71 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (4.2578 + 0.0 + 4.24 + 80.0 + (2.38)) \cdot 138$
 $= 127881.60 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 66025. \text{ Kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1009.6 \cdot 18.0 \cdot 0.49 \cdot 138$
 $= 196829. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 301.3) \cdot (10.0 - 3.0) \cdot 0.7 \cdot 138$
 $= 65267. \text{ Kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 609.6 \cdot 20.0 \cdot 0.74 \cdot 138$
 $= 199424. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 1.0) \cdot 0.74 \cdot 138$
 $= 169510. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:



Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (196829 + 65267) = 262096 \text{ Kg}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (66025 + 199424 + 169510 + 0) = 434960 \text{ Kg}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (196829 + 169510 + 0) = 366339 \text{ Kg}$$

Summary of Failure Path Calculations:

Path 1-1 = 262095 Kg, must exceed W = 0 Kg	or W1 = 124532 Kg
Path 2-2 = 434959 Kg, must exceed W = 0 Kg	or W2 = 10747 Kg
Path 3-3 = 366338 Kg, must exceed W = 0 Kg	or W3 = 127881 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 24.5683 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 342.5683 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.668 %

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 555.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 595.60 mm.

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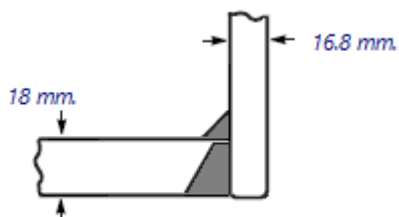

Input, Nozzle Desc: K9-2"
From: 50

Pressure for Reinforcement Calculations	P	6.000	bars
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	3806.00	mm.
Shell Finished (Minimum) Thickness	t	18.0000	mm.
Shell Internal Corrosion Allowance	cas	1.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2600.00	mm.
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	1.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	30.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).


Reinforcement CALCULATION, Description: K9-2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
--	-------	-----



Actual Thickness Used in Calculation

0.660 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)

= $6.0 \cdot 1904.0 / (138 \cdot 1.0 - 0.6 \cdot 6.0)$

= 8.3004 mm.

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]

= $R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)

= $26.4 \cdot (\exp([6.0 / (138.0 \cdot 1.0)] - 1))$

= 0.1150 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 118.3280 mm.

Parallel to Vessel Wall Rn+tn+t 59.1640 mm.

Normal to Vessel Wall (Thickness Limit), no pad Tlnp 39.4100 mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: K9-2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 1.1150 mm.

Wall Thickness per UG16(b), tr16b = 2.5000 mm.

Wall Thickness, shell/head, internal pressure trb1 = 9.3004 mm.

Wall Thickness tb1 = max(trb1, tr16b) = 9.3004 mm.

Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm.

Wall Thickness per table UG-45 tb3 = 5.8000 mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]

= min[5.8, max(9.3004, 2.5)]

= 5.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(1.115, 5.8)

= 5.8000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B**

Govrn. thk, tg = 16.764, tr = 0.115, c = 1.0 mm., E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.007$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -13 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Note: Impact testing of LWN flanges meeting requirements of UCS-66(c)(4) is not required.

Weld Size Calculations, Description: K9-2"



Intermediate Calc. for nozzle/shell Welds Tmin 15.7640 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4672 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 224.8423 mm.

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: